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Chapter 1

Prelimnaries

WORK IN PROGRESS!!!

Thus far, we have been speaking about C^∞ differential forms, but we can define differential forms in more general spaces.

1.1 Function Spaces of Differential Forms

Throughout this section, let $\Omega \subset \mathbb{R}^n$ be an open set and let $0 \leq k \leq n$. Since Ω is an open subset of $\mathbb{R}^n$, we identify $T_p\Omega \simeq \mathbb{R}^n$ for every $p \in \Omega$. Consequently, every $differential$ k-form may be viewed as a map

$$f : \Omega \longrightarrow \Lambda^k(\mathbb{R}^n).$$

With respect to the standard basis of $\Lambda^k(\mathbb{R}^n)$, every k-form admits a unique expansion of the form

$$f = \sum_{I \in \mathscr{T}_k} f^I \, dx^I = \sum_{1 \leq i_1 < \cdots < i_k \leq n} f^{i_1 \cdots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k},$$

where each coefficient $f^I : \Omega \to \mathbb{R}$ is a real-valued function.

Remark 1.1.1. Since $\Lambda^k(\mathbb{R}^n)$ is a finite-dimensional vector space of dimension $\binom{n}{k}$, a differential k-form may equivalently be regarded as a $\binom{n}{k}$-tuple of scalar-valued functions. All regularity notions are defined coefficientwise.

Definition 1.1.2 (L^p differential forms)**.** Let $1 \leq p \leq \infty$. We write

$$f \in L^p(\Omega; \Lambda^k)$$

if and only if every coefficient function satisfies

$$f^I \in L^p(\Omega), \qquad I \in \mathscr{T}_k.$$

Equivalently,

$$L^p(\Omega; \Lambda^k) = L^p(\Omega; \mathbb{R}^{\binom{n}{k}}).$$

The pointwise inner product on $\Lambda^k(\mathbb{R}^n)$ is defined by

$$\langle f, g \rangle = \sum_{I \in \mathscr{T}_k} f^I g^I,$$

with associated pointwise norm

$$|f|^2 = \langle f, f \rangle = \sum_{I \in \mathscr{T}_k} (f^I)^2.$$

The L^p norm of a differential form is obtained by integrating the pointwise norm:

$$\|f\|_{L^p(\Omega;\Lambda^k)} = \left(\int_\Omega |f(x)|^p \, dx \right)^{1/p},$$

with the usual modification when $p = \infty$. In particular,

$$\|f\|^2_{L^2(\Omega;\Lambda^k)} = \int_\Omega \sum_{I \in \mathscr{T}_k} (f^I)^2 \, dx.$$

Definition 1.1.3 (Sobolev differential forms)**.** Let $1 \leq p < \infty$. We write

$$f \in W^{1,p}(\Omega; \Lambda^k)$$

if and only if every coefficient satisfies

$$f^I \in W^{1,p}(\Omega), \qquad I \in \mathscr{I}_k.$$

Thus a differential form belongs to $W^{1,p}$ precisely when all of its coefficients possess weak first-order partial derivatives lying in $L^p(\Omega)$.

For a differential form $f = \sum_I f^I dx^I$, its gradient is defined coefficientwise by

$$\nabla f = \sum_{i=1}^{n} \sum_{I \in \mathscr{I}_k} \frac{\partial f^I}{\partial x^i}\, dx^i \otimes dx^I,$$

and its pointwise norm is

$$|\nabla f|^2 = \sum_{i=1}^{n} \sum_{I \in \mathscr{I}_k} \left(\frac{\partial f^I}{\partial x^i} \right)^2.$$

The Sobolev norm is therefore

$$\|f\|_{W^{1,p}(\Omega;\Lambda^k)} = \left(\|f\|^p_{L^p(\Omega;\Lambda^k)} + \|\nabla f\|^p_{L^p(\Omega;\Lambda^k)} \right)^{1/p},$$

and, in particular,

$$\|f\|^2_{W^{1,2}(\Omega;\Lambda^k)} = \|f\|^2_{L^2(\Omega;\Lambda^k)} + \|\nabla f\|^2_{L^2(\Omega;\Lambda^k)}.$$

Definition 1.1.4 (Classical regularity spaces)**.** For an integer $r \geq 0$, we write

$$f \in C^r(\Omega; \Lambda^k)$$

if and only if every coefficient function belongs to $C^r(\Omega)$.

Similarly,

$$f \in C^{r,\alpha}(\Omega; \Lambda^k), \qquad 0 < \alpha \leq 1,$$

if every coefficient belongs to the Hölder space $C^{r,\alpha}(\Omega)$.

Recall that the Hölder seminorm of a scalar-valued function u is

$$[u]_{C^{0,\alpha}(\Omega)} = \sup_{\substack{x,y \in \Omega \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^\alpha},$$

and the $C^{r,\alpha}$ norm is

$$\|u\|_{C^{r,\alpha}(\Omega)} = \sum_{|\beta| \leq r} \sup_{x \in \Omega} |D^\beta u(x)| + \sum_{|\beta| = r} [D^\beta u]_{C^{0,\alpha}(\Omega)}.$$

The corresponding norm for differential forms is again defined componentwise.

Definition 1.1.5 (Spaces on the closure)**.** We write

$$f \in C^r(\overline{\Omega}; \Lambda^k)$$

if every coefficient extends continuously to $\overline{\Omega}$ together with all derivatives of order at most r.

Likewise,

$$f \in C^{r,\alpha}(\overline{\Omega}; \Lambda^k)$$

if every coefficient extends to a function in $C^{r,\alpha}(\overline{\Omega})$.

Remark 1.1.6. The distinction between $C^{r,\alpha}(\Omega)$ and $C^{r,\alpha}(\overline{\Omega})$ is important. Functions in $C^{r,\alpha}(\Omega)$ are only required to satisfy the Hölder condition in the interior of Ω, whereas elements of $C^{r,\alpha}(\overline{\Omega})$ possess continuous extensions, together with their derivatives up to order r, to the boundary $\partial\Omega$. Boundary conditions such as $\nu \wedge f = 0$ or $\nu \lrcorner f = 0$ are naturally stated in the latter setting.

Remark 1.1.7. Since $\Lambda^k(\mathbb{R}^n)$ is finite dimensional, all norms on $\Lambda^k(\mathbb{R}^n)$ are equivalent. Consequently, the particular choice of Euclidean norm used to define the spaces above is immaterial, and all resulting L^p, Sobolev and Hölder norms are equivalent.

Appendices

Appendix A

Some results from Calculus on $\mathbb{R}^n$

A.1 Results from Differentiation

Theorem A.1.1. ***(Taylor's Theorem)*** *Let $f : U \to \mathbb{R}$ be C^1 in a neighbourhood U of $p \in \mathbb{R}^n$ that is star shaped. Then there exists $g_1, g_2, \cdots g_n$ such that*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i)$$

where $g_i(p) = \frac{\partial}{\partial x^i}|_p f$

Proof. Let $x \in U$. Since U is star shaped, we have $\gamma(t) = p + t(x-p) \in U$ for $t \in [0,1]$. Consider $\frac{d}{dt}(f \circ \gamma(t)) = Df(\gamma(t)) \cdot \gamma'(t) = \sum_{i=1}^{n} \frac{\partial}{\partial x^i} f(\gamma(t))(x^i - p^i)$. Integrate both sides to get

$$\int_{t=0}^{t=1} \frac{d}{dt}(f \circ \gamma(t))dt = f(x) - f(p) = \sum_{i=1}^{n}(x^i - p^i) \int_{t=0}^{t=1} \frac{\partial f}{\partial x^i}(p + t(x-p))dt$$

And let $g_i(x) := \int_{t=0}^{t=1} \frac{\partial f}{\partial x^i}(p + t(x-p))dt$ whence it is easy to verify that $g_i(p) = \frac{\partial f}{\partial x^i}(p)$.

■

Theorem A.1.2. ***(Inverse Function Theorem)*** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Suppose that $\det Df(a) \neq 0$ for a given $a \in \mathbb{R}^n$. Then, there exists U and V open in $\mathbb{R}^n$ such that $a \in U$, $f(a) \in V$ and $f|_U : U \to V$ is a diffeomorphism.*

Theorem A.1.3. ***(Implicit Function Theorem)*** *Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be continuously differentiable in an open set of (a,b) and let $f(a,b) = 0$. Let M be the $m \times m$ matrix*

$$D_{n+j} f^i(a,b); 1 \leq i,j \leq m$$

and suppose $\det M \neq 0$. Then there exists open sets $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^n$ with $a \in A, b \in B$ such that for all $x \in A$ there is a unique $g(x) \in B$ so that $f(x, g(x)) = 0$, and the g so obtained is differentiable.

Proof. Define $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$ given by $F(x,y) := (x, f(x,y))$. This function is clearly differentiable, and moreover,

$$DF = \begin{pmatrix} \frac{\partial x^1}{\partial x^1} & \frac{\partial x^1}{\partial x^2} & \cdots & \frac{\partial x^1}{\partial x^n} & \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^m} \\ \frac{\partial x^2}{\partial x^1} & \frac{\partial x^2}{\partial x^2} & \cdots & \frac{\partial x^2}{\partial x^n} & \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x^n}{\partial x^1} & \frac{\partial x^n}{\partial x^2} & \cdots & \frac{\partial x^n}{\partial x^n} & \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^m} \\ \frac{\partial f^1}{\partial x^1} & \frac{\partial f^1}{\partial x^2} & \cdots & \frac{\partial f^1}{\partial x^n} & \frac{\partial f^1}{\partial y^1} & \frac{\partial f^1}{\partial y^2} & \cdots & \frac{\partial f^1}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} & \frac{\partial f^m}{\partial x^2} & \cdots & \frac{\partial f^m}{\partial x^n} & \frac{\partial f^m}{\partial y^1} & \frac{\partial f^m}{\partial y^2} & \cdots & \frac{\partial f^m}{\partial y^m} \end{pmatrix}$$

And it is easy to see that $\det DF(a,b) = detM \neq 0$. Therefore, from Inverse function theorem (with litte modification), there exists an open set $A \times B$ of $\mathbb{R}^n \times \mathbb{R}^m$, and an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ such that $(a,b) \in A \times B$, $(a,0) \in W$ and $F : A \times B \to W$ is a diffeomorphism. $F^{-1} : W \to A \times B$ is also a diffeomorphism. Moreover, note that $F^{-1}(p,q)$ is of the form $(p, k(p,q))$ for some differentiable function $k : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$. $\pi \circ F(x,y) = \pi \circ (x, f(x,y)) = f(x,y)$ so $\pi \circ F \equiv f$. Look at $f \circ (p, k(p,q)) = f \circ F^{-1}(p,q) = \pi(p,q) = q$ so $f \circ (p, k(p,0)) = 0$. Our $g(p)$ is given by $k(p,0)$, and it is differentiable as a composition of differentiable functions.

■

Theorem A.1.4. ***(A corollary of A.1.3)*** *Let $f : \mathbb{R}^n \to \mathbb{R}^p (p \leq n)$ be continuously differentiable in an open set containing $a \in \mathbb{R}^n$. If $f(a) = 0$ and the matrix $D_j(f^i(a))_{p \times n}$ has rank p, then there exists an open neighbourhood A of a and a diffeomorphism $h : A \to h(A) \subset \mathbb{R}^n$ such that $f \circ h(x_1, x_2 \cdots x_n) = (x_{n-p+1}, x_{n-p+2} \cdots, x_n)$.*

Proof. Consider $f : \mathbb{R}^{n-p} \times \mathbb{R}^p \to \mathbb{R}^p$. Suppose that the matrix $Q = D_{n-p+i}(f^j); i,j \leq p$ be invertible ($\det Q \neq 0$). Let, as before, $F : \mathbb{R}^{n-p} \times \mathbb{R}^p \to \mathbb{R}^{n-p} \times \mathbb{R}^p$ given by

$$F(p,q) = (p, f(p,q))$$

Since $\det DF \neq 0$, there exists, around $a = (a^1, a^2, \cdots, a^{n-p}, a^{n-p+1}, \cdots a^n)$ a neighbourhood $A_{n-p} \times A_p \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ and a neighbourhood $W \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ containing $F(a) = (a_1, a_2 \cdots a_{n-p}, 0)$ such that $F : A_{n-p} \times A_p \to W$ is invertible and whose inverse is continuously differentiable. Note that, the inverse of F ought to be of the form $F^{-1}(x_1, x_2, \cdots, x_{n-p}, y_{n-p+1}, \cdots, y_n) = (\vec{x}, b(\vec{x}, \vec{y}))$ (let $\vec{x} = (x_1, x_2, \cdots x_{n-p}) \in \mathbb{R}^{n-p}$ and $\vec{y} = (y_{n-p+1}, y_{n-p+2}, \cdots, y_n) \in \mathbb{R}^p$) where it is understood that $b : \mathbb{R}^n \to \mathbb{R}^p$ is continuously differentiable. Also note that $\pi_2 \circ F \equiv f$. We have that

$$f \circ F^{-1}(x_1, x_2 \cdots x_{n-p}, x_{n-p+1}, \cdots x_n) =$$

$$\pi_2(x_1, x_2, \cdots x_{n-p}, x_{n-p+1}, \cdots x_n) = (x_{n-p+1}, x_{n-p+2}, \cdots, x_n)$$

Case 2: If the indices where the matrix $\{(\partial_i f^j(a)); 1 \leq j \leq p, i = i_1, i_2, \cdots i_p\}$ is not the last p indices, we write the linear transformation $\text{Swap} : \mathbb{R}^n \to \mathbb{R}^n$ that takes these p indices to the last p indices. This is a linear transformation that is invertible since it takes basis elements to basis elements. If we look then, at $f \circ \text{Swap} : \mathbb{R}^n \to \mathbb{R}^p$, we see that this is a function of the form of Case 1, so we have a h such that $f \circ \text{Swap} \circ h(x_1, x_2, \cdots x_n) = (x_{n-p+1}, x_{n-p+2}, \cdots x_n)$ which means our desired function is $\text{Swap} \circ h$.

■

Appendix B

Differential Forms

B.1 Exterior Algebra: Definitions and Results

Definition B.1.1. (Dual Space V^*) Let V be a vector space over $\mathbb{R}$. The set of all linear maps $f : V \to \mathbb{R}$ is called the *dual space* of V and is denoted by V^* or $\Lambda^1(V)$.

Lemma B.1.2. *Let $e_1, e_2, \cdots, e_n$ be a basis for V. Then the vectors $\xi^1, \xi^2, \cdots, \xi^n \in \Lambda^1(V)$ defined by $\xi^i(e_j) = \delta_{ij}$ forms a basis for $\Lambda^1(V)$.*

Definition B.1.3. (Multilinear Maps (Tensors))

$$f : \underbrace{V \times V \times \cdots \times V}_{k \text{ copies}} \to \mathbb{R}$$

is a *k-tensor* if it is linear in each coordinate.

Note: The space of all tensors forms a vector space.

Lemma B.1.4. *For the space of all $k-$tensors from V, $\{dx^{i_1} \otimes dx^{i_2} \otimes \cdots \otimes dx^{i_k} : (i_1, i_2, \cdots, i_k) \in \{1, 2, \cdots, n\}^k\}$ forms a basis (Here we have adopted the notational choice of writing dx^i in place ξ^i).*

Definition B.1.5. (σf) Let f be a $k-$ tensor and $\sigma \in Sym(k)$. We define

$$\sigma f(v_1, v_2, \cdots, v_k) := f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$$

B.1.1 σf as a group action

Let $X =$ set of all $k-$ tensors. Define $\rho : sym(k) \times X \to X$ by $\rho(\sigma, f) := \sigma f$. We see that ρ is a group action since

1. $\rho(1, f) = f$
2. $\rho(\sigma_1, \rho(\sigma_2, f)) = \rho(\sigma_1 \sigma_2, f)$

Definition B.1.6. (Alternating Tensors) A $k-$tensor is alternating if for every $\sigma \in sym(k)$,

$$f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)}) = \text{sign}(\sigma) f(v_1, v_2, \cdots, v_k)$$

or

$$\sigma f \equiv \text{sign}(\sigma) f$$

Definition B.1.7. (Symmetric Tensors) A $k-$tensor is symmetric if for every $\sigma \in sym(k)$,

$$f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)}) = f(v_1, v_2, \cdots, v_k)$$

or

$$\sigma f \equiv f$$

Definition B.1.8. (Symmetrization Operator Sf)

$$S : \{k - \text{tensors}\} \to \{k - \text{tensors}\}$$

given by $Sf(v_1, v_2, \cdots, v_k) = \sum_{\sigma \in sym(k)} f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$ or

$$Sf \equiv \sum_{\sigma \in sym(k)} \sigma f$$

is called the *symmetrization operator*

Definition B.1.9. (Alternation or Antisymmetrizer Operator Af)

$$A : \{k - \text{tensors}\} \to \{k - \text{tensors}\}$$

given by $Af(v_1, v_2, \cdots, v_k) = \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f(v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(k)})$ or

$$Af \equiv \sum_{\sigma \in sym(k)} \text{sign}(\sigma) \sigma f$$

is called the *alternation operator*

Theorem B.1.10. *If f is a $k-$tensor, then*

1. *Sf is a symmetric $k-$tensor.*
2. *Af is an alternating $k-$tensor.*

Proof. 1) $Sf := \sum_{\sigma \in sym(k)} \sigma f$. Let $\tau \in sym(k)$.

$$\tau Sf = \sum_{\sigma \in sym(k)} (\tau\sigma) f = Sf$$

(since $\tau\sigma$ runs over all of $sym(k)$ once and exactly once).

2) $Af := \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f$. Let $\tau \in sym(k)$.

$$\tau Af = \sum_{\sigma \in sym(k)} \text{sign}(\sigma\tau)\,\text{sign}(\tau)(\tau\sigma) f =$$

$$\text{sign}(\tau) Af$$

.

■

Lemma B.1.11.

1. *If f is an alternating $k-$ tensor ($\sigma f = \text{sign}(\sigma) f$) then $Af = k! f$.*
2. *If f is an alternating $k-$ tensor then $Sf = 0$.*
3. *If f is a symmetric $k-$tensor ($\sigma f = f$) then $Af = 0$.*
4. *If f is a symmetric $k-$tensor then $Sf = k! f$*

Proof. 1)

$$Af := \sum_{\sigma \in sym(k)} \text{sign}(\sigma)\,\text{sign}(\sigma) f = \sum_{\sigma \in sym(k)} f = k! f$$

2) If f is alternating, then

$$Sf = \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f = 0$$

3) If f is symmetric, then

$$A(f) := \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f = 0$$

4) If f is symmetric, then

$$Sf := \sum_{\sigma \in sym(k)} \sigma f = k! f$$

■

B.1.2 Tensor Product, Alternating Forms and the Wedge Product

Definition B.1.12. (Tensor Product) Let f be a $k-$tensor and g be a $l-$tensor. We define the **tensor product** $f \otimes g$ to be the $(k+l)-$tensor given by

$$f \otimes g(v_1, v_2, \cdots, v_k, v_{k+1}, v_{k+2}, \cdots, v_{k+l}) := f(v_1, v_2, \cdots, v_k) \cdot g(v_{k+1}, v_{k+2}, \cdots, v_{k+l})$$

Note: The tensor product is *bilinear* (linear in each argument) and *associative* ($(f \otimes g) \otimes h = f \otimes (g \otimes h)$) but not *commutative*.

Definition B.1.13. (Wedge Product) Let f be an alternating $k-$tensor ($f \in \Lambda^k(V)$) and g an alternating $l-$tensor ($g \in \Lambda^l(V)$). We define

$$f \wedge g := \frac{1}{k!l!} A(f \otimes g)$$

or written explicitly

$$f \wedge g(v_1, \cdots, v_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in sym(k+l)} \text{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

Definition B.1.14. **($Sym(k,l)$)** Given natural numbers k and l, we define $Sym(k,l)$ as follows:

$$Sym(k,l) := \{\sigma \in sym(k+l) : \sigma(1) < \sigma(2) < \cdots < \sigma(k) \text{ and } \sigma(k+1) < \sigma(k+2) < \cdots < \sigma(k+l)\}$$

Theorem B.1.15. ***(Equivalent Expressions for the Wedge Product)*** *The two expressions*

1. $$f \wedge g(v_1, \cdots, v_{k+l}) := \frac{1}{k!l!} \sum_{\sigma \in sym(k+l)} \text{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

2. $$f \wedge g(v_1, \cdots, v_{l+l}) = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

are equivalent.

Proof. Consider $G = sym(k+l)$ and consider the subgroup

$$H := \{\sigma \in sym(k+l) : \sigma|_{\{1,\cdots,k\}} \in \{1, \cdots, k\} \text{ and } \sigma|_{\{k+1,\cdots,l+l\}} \in \{k+1, \cdots, k+l\}\}$$

(One can view every element $\sigma \in H$ as $\pi_1 \times \pi_2$ where $\pi_1 \in sym\{1, \cdots, k\}$ and $\pi_2 \in sym\{k+1, \cdots, k_l\}$).

Let $\alpha, \beta \in G$ and define an equivalence relation $\alpha \sim \beta$ if and only if there exists $h \in H$ such that $\alpha h = \beta$

Let S_σ be the expression $\text{sign}(\sigma)\sigma(f \otimes g)$.

Suppose $\alpha \sim \beta$ or $\alpha h = \beta$ where $h = \pi_1 \times \pi_2$.

$$S_\alpha = \text{sign}(\alpha) f(v_{\alpha(1)}, \cdots, v_{\alpha(k)}) g(v_{\alpha(k+1)}, \cdots, v_{\alpha(k+l)})$$

and

$$S_\beta = \text{sign}(\beta) f(v_{\beta(1)}, \cdots, v_{\beta(k)}) g(v_{\beta(k+1)}, \cdots, v_{\beta(k+l)})$$

which is equal to

$$\text{sign}(\alpha)\,\text{sign}(\pi_1)\,\text{sign}(\pi_2) f(v_{\alpha(h(1))}, \cdots, v_{\alpha(h(k))}) g(v_{\alpha(h(k+1))}, \cdots, v_{\alpha(h(k+l))})$$

$$= \text{sign}(\alpha)\,\text{sign}(\pi_1)^2\,\text{sign}(\pi_2)^2 f(v_{\alpha(1)}, \cdots, v_{\alpha(k)}) g(v_{\alpha(k+1)}, \cdots, v_{\alpha(k+l)}) = S_\alpha$$

So whenever $\alpha \sim \beta$, $S_\alpha = S_\beta$.

We can also see that, via rearrangement, for every $\sigma \in sym(k+l)$ there exists an $\alpha \in sym(k,l)$ and a $h \in H$ such that $\sigma h = \alpha$ or $\sigma \sim \alpha$.

The first expression can now be written as

$$\frac{1}{k!l!} \sum_{\sigma \in sym(k,l)} \sum_{\alpha \in [\sigma]} S_\alpha$$

but for elements of the same equivalence class, S_α are all the same and is equal to S_σ, so we have

$$\frac{1}{k!l!} \sum_{\sigma \in sym(k,l)} S_\sigma (k!l!) = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma) \sigma(f \otimes g)$$

(The $k!l!$ comes from the fact that $|H| = k!l!$ and so the size of each coset is $k!l!$) and the equality is proved.

■

Theorem B.1.16. *For $f \in \Lambda^k(V)$ and $g \in \Lambda^l(V)$*

$$f \wedge g = (-1)^{kl} g \wedge f$$

Proof.

$$f \wedge g = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma) \sigma(f \otimes g)$$

Let τ be the element of $sym(k+l)$ given by $\tau(1) = k+1, \tau(2) = k+2, \cdots \tau(l) = k+l$ and $\tau(l+1) = 1, \tau(l+2) = 2, \cdots \tau(l+k) = k$ (We are moving the block $\{k+1, k+2, \cdots, k+l\}$ back, behind the block $\{1, 2 \cdots k\}$, which amounts to a total of kl swaps/transpositions). Note that τ acts as a correspondence between $sym(k,l)$ and $sym(l,k)$ since for every $\sigma \in sym(k,l)$, we have $\sigma(\tau(l+1)) < \sigma(\tau(l+2)) < \cdots < \sigma(\tau(l+k))$ and $\sigma(\tau(1)) < \sigma(\tau(2)) < \cdots < \sigma(\tau(l))$ so $\sigma(\tau)$ is in $sym(l,k)$.

$$f \wedge g(v_1, \cdots, v_{k+l}) = \sum_{\sigma \in sym(k,l)} \text{sign}(\sigma) f(v_{\sigma(1)}, \cdots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)})$$

$$= \sum_{\sigma \in sym(k,l)} \text{sign}(\tau\sigma)\,\text{sign}(\tau) g(v_{\sigma(\tau(1))}, \cdots, v_{\sigma(\tau(l))}) f(v_{\sigma(\tau(l+1))}, \cdots, v_{\sigma(\tau(l+k))})$$

$$= \sum_{\gamma \in sym(l,k)} \text{sign}(\gamma)(-1)^{kl} g(v_{\gamma(1)}, \cdots, v_{\gamma(l)}) f(v_{\gamma(l+1)}, \cdots, v_{\gamma(l+k)}) = (-1)^{kl} g \wedge f$$

■

B.1.3 The Associativity of the Wedge Product

Lemma B.1.17. *Let $f \in \Lambda^k(V)$ and $g \in \Lambda^l(V)$, then:*

1. $A(Af \otimes G) = k!A(f \otimes g)$
2. $A(f \otimes Ag) = l!A(f \otimes g)$

Proof. 1)

$$A(Af \otimes g) = \sum_{\sigma \in sym(k+l)} \text{sign}(\sigma)\sigma(Af \otimes g)$$

$$Af := \sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ preserves } \{k+1,\cdots,k+l\}}} \text{sign}(\tau)\tau(f)$$

and

$$A(Af \otimes g) := \sum_{\sigma \in sym(k+l)} \text{sign}(\sigma)(Af(X_{\sigma(1)}, \cdots, X_{\sigma(k)})g(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)}))$$

$$= \sum_{\sigma \in sym(k+l)} \text{sign}(\sigma)\Big(\sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ preserves} \\ \{k+1,\cdots,k+l\}}} \text{sign}(\tau) f(X_{\sigma\tau(1)}, \cdots, X_{\sigma\tau(k)})\Big) g(X_{\sigma(k+1)}, \cdots, X_{\sigma(k+l)}))$$

$$= \sum_{\sigma \in sym(k+l)} \sum_{\substack{\tau \in sym(k+l) \\ \tau \text{ fixes} \\ \{k+1,\cdots,k+l\}}} \text{sign}(\sigma\tau) f(X_{\sigma\tau(1)}, \cdots, X_{\sigma\tau(k)}) g(X_{\sigma\tau(k+1)}, \cdots, X_{\sigma\tau(k+l)})$$

The number of such τ that fixes $\{k+1, \cdots, k+l\}$ are precisely $k!$, and for each such τ, $\sigma\tau$ runs over all of $sym(k+l)$ once and exactly once. Hence we have

$$A(Af \otimes g) = k! \sum_{\gamma \in sym(k+l)} \text{sign}(\gamma)\gamma(f \otimes g) = k!A(f \otimes g)$$

2) The 2nd part is proved similarly.

■

Theorem B.1.18. ***(Associativity of Wedge Product)*** *For any $f \in \Lambda^k(V), g \in \Lambda^l(V), h \in \Lambda^m(V)$*

$$f \wedge (g \wedge h) = (f \wedge g) \wedge h$$

Proof.

$$f \wedge (g \wedge h) := \frac{1}{k!(l+m)!} A(f \otimes (g \wedge h)) = \frac{1}{k!(l+m)!} A(f \otimes (\frac{1}{l!m!} A(g \otimes h)))$$

$$= \frac{1}{k!l!m!(l+m)!} A(f \otimes A(g \otimes h)) = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$$

and likewise

$$(f \wedge g) \wedge h = \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)k!l!m!} A(A(f \otimes g) \otimes h)$$

$$= \frac{1}{k!l!m!}(k+l)!A(f \otimes g \otimes h) = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$$

■

Note: *The proof of B.1.18 shows that if f_{l_1} is an l_1-form, f_{l_2} is an l_2-form and so on, $f_{l_1} \wedge f_{l_2} \wedge \cdots \wedge f_{l_k}$ is given by*

$$f_{l_1} \wedge f_{l_2} \wedge \cdots \wedge f_{l_k} = \frac{1}{l_1!l_2! \cdots l_k!} A(f_{l_1} \otimes f_{l_2} \otimes \cdots \otimes f_{l_k}) \tag{B.1.1}$$

Short Digression on Determinants

Definition B.1.19. (Determinant (*Leibniz Formula*)) Let $\{a_{ij} : 1 \leq i, j \leq n\}$ be a matrix. We define determinant of A as

$$\det(A) := \sum_{\sigma \in sym(n)} \text{sign}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}$$

(To be thought of as multiplying one element from each row chosen from distinct columns, and taking into account the sign of the permutation of the columns).

Definition B.1.20. (Minor of $(i,j)-$entry) Let A be a matrix and let a_{ij} be the $(i,j)-$entry of A. The *minor* of a_{ij} is the determinant of the matrix obtained by deleting the ith row and j^{th} column from A. It is denoted m_{ij}. The matrix itself (after deleting i^{th} row and j^{th} column) is denoted M_{ij}.

Definition B.1.21. (Laplace's Expansion) Fix a row i. Then the Laplace's expansion gives

$$\det(B) = \sum_{j=1}^{n} (-1)^{i+j} b_{ij} \det(M_{ij})$$

Note: From

$$\sum_{\sigma \in sym(n)} \text{sign}(\sigma) \prod_{q=1}^{n} a_{q,\sigma(q)} = \sum_{j=1}^{n} \sum_{\substack{\sigma \in sym(n) \\ \sigma : i \mapsto j}} \text{sign}(\sigma) a_{i,j} \prod_{\substack{q=1 \\ q \neq i}}^{n} a_{q,\sigma(q)} \tag{B.1.2}$$

we can see that the Laplace's expansion is a consequence of the Leibniz formula for determinants.

Theorem B.1.22. ***(Wedge Product of $1-$forms or Functionals)*** *Let $f_1, f_2, \cdots, f_k$ be in $\Lambda^1(V) \equiv V^*$. Then*

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = det(f_i(x_j))$$

Proof. From B.1.1, we have

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, \cdots, x_k) = A(f_1 \otimes f_2 \otimes \cdots \otimes f_k)$$

or

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \sum_{\sigma \in sym(k)} \text{sign}(\sigma) f_1(x_{\sigma(1)}) f_2(x_{\sigma(2)}) \cdots f_k(x_{\sigma(k)})$$

which is precisely the Leibniz formula for determinants. Hence

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \det(f_i(x_j))$$

■

Lemma B.1.23. *Any collection $f_1, f_2, \cdots, f_k$ of $1-$forms is linearly independent if and only if $f_1 \wedge f_2 \wedge \cdots \wedge f_k \neq 0$*

Proof. $\implies$) Say $f_1, \cdots, f_k$ is a linearly independent collection of functionals. Then it is part of a larger collection that forms a basis for V^*. Let $x_1, x_2, \cdots, x_k$ be the vectors in V such that $f_i(x_j) = \delta_{ij}$. Then

$$f_1 \wedge f_2 \wedge \cdots \wedge f_k(x_1, x_2, \cdots, x_k) = \det(\delta_{ij}) = 1$$

$\impliedby$) Let $f_1, f_2, \cdots, f_k$ be linearly dependent. Then (without loss of generality) $f_1 = c_2 f_2 + c_3 f_3 + \cdots c_k f_k$ and $f_1 \wedge f_2 \wedge \cdots \wedge f_k = c_2(f_2 \wedge f_2 \wedge f_3 \cdots \wedge f_k) + c_3(f_3 \wedge f_2 \wedge f_3 \wedge \cdots \wedge f_k) + \cdots c_k(f_k \wedge f_2 \wedge f_3 \wedge \cdots \wedge f_k) = 0$

■

B.1.4 Dimension of Alternating k-forms

Let $E_1, E_2, \cdots, E_n$ be a basis for V. Let $\xi^1, \xi^2, \cdots, \xi^n \in V^* = \Lambda^k(V)$ be the dual basis to $E_1, \cdots E_n$.

Theorem B.1.24. ***(Basis for** $\Lambda^k(V)$**)*** *The set*

$$\{\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} : 1 \leq i_1 < i_2 < \cdots < i_k \leq n\}$$

forms a basis for $\Lambda^k(V)$.

Proof. **Linear Independence:** Look at

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} \xi^{i_1} \wedge xi^{i_2} \wedge \cdots \wedge \xi^{i_k} \equiv 0$$

in $\Lambda^k(V)$. Hence,

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} \xi^{i_1} \wedge xi^{i_2} \wedge \cdots \wedge \xi^{i_k}(E_{j_1}, E_{j_2}, \cdots, E_{j_k}) =$$

$$\sum_{1 \leq i_1 < \cdots < i_k \leq n} c_{i_1 i_2 \cdots i_k} det(\xi^{i_r}(E_{j_s})) = 0$$

$$\implies c_{j_1 j_2 \cdots j_k} = 0$$

for every increasing k-tuple $j_1, j_2, \cdots, j_k$ and hence $\{\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} : 1 \leq i_1 < \cdots < i_k \leq n\}$ is linearly independent.

Span: Let $f \in \Lambda^k(V)$. Let $x_r = \sum_{j_1=1}^n \xi^{j_1}(x_r) E_{j_1}$ for $r = 1, 2, \cdots, k$.

$$f(x_1, x_2, \cdots, x_k) = f(\sum_{j_1=1}^n \xi^{j_1}(x_1) E_{j_1}, \sum_{j_1=1}^n \xi^{j_1}(x_2) E_{j_1}, \cdots, \sum_{j_1=1}^n \xi^{j_1}(x_k) E_{j_1})$$

$$= \sum_{j_1=1}^n \sum_{j_2=1}^n \cdots \sum_{j_k=1}^n \xi^{j_1}(x_1) \xi^{j_2}(x_2) \cdots \xi^{j_k}(x_k) f(E_{j_1}, E_{j_2}, \cdots, E_{j_k})$$

If $j_r = j_s$ for some r and s, note that the summand is 0. Hence we rewrite the above sum as:

$$\sum_{\substack{1 \leq j_1, j_2, \cdots, j_k \leq n \\ j_r \text{ distinct}}} \xi^{j_1}(x_1) \xi^{j_2}(x_2) \cdots \xi^{j_k}(x_k) f(E_{j_1}, E_{j_2}, \cdots, E_{j_k})$$

We now note that each choice of $1 \leq j_1, j_2, \cdots, j_k \leq n$ tuple is some $\sigma \in sym(k)$ of the ordered $1 \leq j_1 < j_2 < \cdots < j_k \leq n$ (Simply reorder the elements $j_1, j_2, \cdots, j_k$ such that we achieve $\sigma(j_1) < \sigma(j_2) < \cdots < \sigma(j_k)$), and hence the above sum can be rewritten as:

$$\sum_{1 \leq j_1 < \cdots < j_k \leq n} \sum_{\sigma \in sym(k)} \xi^{j_{\sigma(1)}}(x_1) \xi^{j_{\sigma(2)}}(x_2) \cdots \xi^{j_{\sigma(k)}}(x_k) f(E_{j_{\sigma(1)}}, \cdots, E_{j_{\sigma(k)}})$$

which is precisely

$$\sum_{1 \leq j_1 < \cdots < j_k \leq n} f(E_{j_1}, \cdots, E_{j_k}) \sum_{\sigma \in sym(k)} \text{sign}(\sigma) \xi^{j_{\sigma(1)}}(x_1) \xi^{j_{\sigma(2)}}(x_2) \cdots \xi^{j_{\sigma(k)}}(x_k)$$

$$= \sum_{1 \leq j_1 < \cdots < j_k \leq n} f(E_{j_1}, \cdots, E_{j_k}) \xi^{j_1} \wedge \xi^{j_2} \cdots \wedge \xi^{j_k}(x_1, x_2, \cdots, x_k)$$

which means f can be written as a linear combination of $\xi^{j_1} \wedge \cdots \wedge \xi^{j_k}$.

■

Lemma B.1.25. *Let* $\beta^i \in \Lambda^1(V)$ *for* $1 \leq i \leq k$ *and let* $\gamma^i \in \Lambda^1(V)$ *and if*

$$\beta^i = \sum_{j=1}^k \alpha_j^i \gamma^j$$

then

$$\beta^1 \wedge \beta^2 \wedge \cdots \wedge \beta^k = \det(\alpha^i_j)\gamma^1 \wedge \gamma^2 \wedge \cdots \gamma^k$$

Proof.

$$\begin{aligned}\beta^1 \wedge \cdots \wedge \beta^k &= \sum_{j_1=1}^{k}\sum_{j_2=1}^{k}\cdots\sum_{j_k=1}^{k}(\alpha^1_{j_1}\alpha^2_{j_2}\cdots\alpha^k_{j_k})(\gamma^{j_1}\wedge\gamma^{j_2}\wedge\cdots\wedge\gamma^{j_k}) \\ &= \sum_{\substack{1\le j_1<\cdots<j_k\le k \\ j_r \text{ distinct}}}(\alpha^1_{j_1}\alpha^2_{j_2}\cdots\alpha^k_{j_k})\gamma^{j_1}\wedge\gamma^{j_2}\cdots\wedge\gamma^{j_k} \\ &= \sum_{1\le j_1<\cdots<j_k\le k}\ \sum_{\sigma\in sym(k)}\operatorname{sign}(\sigma)\alpha^1_{j_1}\cdots\alpha^k_{j_k}\gamma^1\wedge\gamma^2\cdots\gamma^k \\ &= \det(\alpha^i_j)\gamma^1\wedge\gamma^2\wedge\cdots\wedge\gamma^k\end{aligned}$$

■

Lemma B.1.26. *Let $f \in \Lambda^k(V)$ and let $u_1, u_2, \cdots, u_k$ and $v_1, v_2, \cdots, v_k \in V$ be related by*

$$u_i = \sum_{j=1}^{n} a^i_j v_j$$

Then

$$f(u_1, u_2, \cdots, u_k) = \det(a^i_j) f(v_1, v_2, \cdots, v_k)$$

Proof. $f(\sum_{j_1=1}^n a^1_{j_1}v_{j_1}, \sum_{j_2=1}^k a^2_{j_2}v_{j_k}, \cdots, \sum_{j_k=1}^n a^k_{j_k}v_j)$

$$= \sum_{\substack{j_1,j_2,\cdots,j_k \\ each distinct}} a^1_{j_1}a^2_{j_2}\cdots a^k_{j_k} f(v_{j_1}, v_{j_2}, \cdots, v_{j_k})$$

$$= \sum_{\substack{1\le j_1<j_2<\cdots<j_k\le k \\ each distinct}}\ \sum_{\sigma\in sym(k)}\operatorname{sign}(\sigma)a^1_{\sigma(1)}a^2(\sigma(2))\cdots a^k_{\sigma(k)}f(v_1, v_2, \cdots, v_k) =$$

$$\det(a^i_j)f(v_1, v_2, \cdots, v_k)$$

■

Notation: Let $\mathscr{T}_k$ denote the set

$$\mathscr{T}_k := \{I \equiv (i_1, i_2, \cdots, i_k) : 1 \le i_1 < i_2 < \cdots < i_k \le n\} \tag{B.1.3}$$

Suppose that $E_1, E_2, \cdots, E_n$ are the basis for V, and $\xi^1, \xi^2, \cdots, \xi^n$ be the dual basis for $\Lambda^1(V)$. Then, for $I = (i_1, i_2, \cdots, i_k)$ we have the notation

$$\xi^I := \xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} \tag{B.1.4}$$

and hence for any $k-$form $f : V \times V \times \cdots \times V \to \mathbb{R}$ we have the notation for the expansion

$$f := \sum_{I\in\mathscr{T}_k} f^I \xi^I \tag{B.1.5}$$

B.2 Tangent vectors in $\mathbb{R}^n$ and Vector Fields

We denote points in $\mathbb{R}^n$ by

$$p = (p^1, p^2, \cdots, p^n)$$

and points in $T_p(\mathbb{R}^n)$, the tangent space of $\mathbb{R}^n$ at p (the space of all vectors v eminating from p), by

$$v = [v^1, v^2, \cdots, v^n]^T$$

B.2.1 The Directional Derivative and Derivations

Definition B.2.1. (Directional Derivative) Let $p \in \mathbb{R}^n$ be given and let $f \in C_p^\infty$, the space of all functions $f : U \to \mathbb{R}$ such that $p \in U$ and f is C^∞ in U. Let $v \in T_p(\mathbb{R}^n)$. Let $\gamma(t) := p + tv$ and look at

$$\frac{d}{dt}|_{t=0}(f \circ \gamma(t)) = Df(\gamma(0)) \cdot \gamma'(0) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}|_p f$$

If we fix a point p (arbitrary) and a direction $v \in T_p(\mathbb{R}^n)$, we have an operator $D_{p,v} : C_p^\infty \to \mathbb{R}$ given by

$$D_{p,v}(f) := \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\Big|_p (f)$$

This operator is *Linear* and follows the property

$$D_{p,v}(f \cdot g) = (D_{p,v}(f))g(p) + (D_{p,v}(g))f(p)$$

The operator $D_{p,v}$ is called the ***directional derivative*** *at p in the direction v.*

Definition B.2.2. (Germs of Functions) Fix a point p and look at all pairs (f, U) where $f : U \to \mathbb{R}$, $p \in U \subseteq \mathbb{R}^n$ open and $f \in C^\infty(U)$. We say $(f, U) \sim (g, V)$ if f and g agree on an open subset of $U \cap V$. This is an equivalence relation and we (by abuse of notation) denote the equivalence class of (f, U) as simply f. The set of all such equivalence classes is denoted by C_p^∞ known as the ***Germs of functions at*** p.

Defining addition, scalar multiplication and function multiplication on C_p^∞ by the usual operations on representatives, we see that C_p^∞ is a commutative ring with identity. Moreover, it is also a Vector space over $\mathbb{R}$.

Definition B.2.3. (Point Derivation) Fix a point $p \in \mathbb{R}^n$. We say $D : C_p^\infty \to \mathbb{R}$ is a *point derivation* (at p) if:

1. D is linear.
2. D obeys *Leibniz's Rule*:
$$D(f \cdot g) = D(f)g(p) + D(g)f(p)$$

and the space of all point derivations at p is denoted by $\mathscr{D}_p(\mathbb{R}^n)$

Lemma B.2.4. *If $D \in \mathscr{D}_p(\mathbb{R}^n)$, then $D(c) = 0$ for every constant function $c \in C_p^\infty$.*

Proof. $D(1) = D(1 \cdot 1) = 1D(1) + 1D(1) = 2D(1)$ which implies $D(1) = 0$. From linearity, we have the result.

■

Lemma B.2.5. *The space of all point derivations $\mathscr{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$ and $T_p(\mathbb{R}^n) \cong \mathscr{D}_p(\mathbb{R}^n)$. Moreover, $D_{p,()} : T_p(\mathbb{R}^n) \to \mathscr{D}_p(\mathbb{R}^n)$ given by $v \mapsto D_{p,v}$ is the isomorphism.*

Proof. That $\mathscr{D}_p(\mathbb{R}^n)$ is a vector space is clear. Look at the map $v \mapsto D_{p,v}$. $v + cw \mapsto D_{p,v+cw}$ Where

$$D_{p,v+cw}(f) = \sum_{i=1}^{n} (v^i + cw^i) \frac{\partial}{\partial x^i}\Big|_p (f) = D_{p,v}(f) + cD_{p,w}(f)$$

Which makes $v \mapsto D_{p,v}$ linear.

Injectivity: Let $D_{p,v}(f) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\big|_p (f) \equiv 0$ for all $f \in C_p^\infty$ Let $f = x^j : \mathbb{R}^n \to \mathbb{R}$ that picks out the $j-$th coordinate. Then $D_{p,v}(x^j) = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\big|_p (x^j) = v^j = 0$. Therefore, $v = 0$ and hence, the nullspace is trivial.

Surjectivity: Let $f \in C_p^\infty$. We have $f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i)$ where $g^i(p) = \frac{\partial f}{\partial x^i}(p)$

(From A.1.1)

Let $D \in \mathscr{D}_p(\mathbb{R}^n)$.

$$D(f(x)) = D(f(p)) + \sum_{i=1}^{n} D(g_i(x) \cdot (x^i - p^i))$$

$$= 0 + \sum_{i=1}^{n} D(g_i(x))(0) + D(x^i - p^i)\frac{\partial f}{\partial x^i}(p) = \sum_{i=1}^{n} D(x^i)\frac{\partial}{\partial x^i}\big|_p f$$

Which means

$$D \equiv \sum_{i=1}^{n} D(x^i)\frac{\partial}{\partial x^i}\big|_p \equiv D_{p,<D(x^1),D(x^2)\cdots,D(x^n)>}$$

And hence, the map $v \mapsto D_{p,v}$ is surjective.

■

Definition B.2.6. (Vector Field) A *vector field* on $\mathbb{R}^n$ is a map $X : U \mapsto \amalg_{p\in U} T_p(U)$. For every $p \in U$, $X(p)$ is a vector in the tangent space at p.

$$X(p) = \sum_{i=1}^{n} a^i(p)\frac{\partial}{\partial x^i}\big|_p$$

We call X a smooth vector field if $a^i : U \to \mathbb{R}$ are C^∞.

Let $f \in C^\infty_p$, that is $f : U \to \mathbb{R}$, a vector field X to f is

$$Xf(p) := \sum a_i(p)\frac{\partial}{\partial x^i}|_p f$$

Then

$$Xf : U \to \mathbb{R}$$

is again a C^∞ function.

B.3 Differential Forms

We start with defining what are $1-$forms.

Definition B.3.1. ($1-$forms) A function

$$\omega : U \to \amalg_{p\in U} \Lambda^1(T_p(\mathbb{R}^n))$$

such that $\omega(p) \in \Lambda^1(T_p(\mathbb{R}^n)) \equiv T^*_p(\mathbb{R}^n)$ is called a $1-$form.

Note: Let $f : U \to \mathbb{R}$ be any C^∞ function. We can construct a $1-$form from f as follows:

$$df : U \to \amalg_{p\in U} \Lambda^1(T_p(U))$$

such that

$$df_p \in \Lambda^1(T_p(U))$$
$$df_p : V \to \mathbb{R}$$

given by the action (for $v = \sum_{j=1}^n v^j \frac{\partial}{\partial x^j}|_p \in T_p(U)$):

$$df_p(v) = df_p(\sum_j v^j \frac{\partial}{\partial x^j}|_p) = \sum_j v^j \frac{\partial}{\partial x^j}|_p(f) \in \mathbb{R}$$

Suppose that $v \in T_p(\mathbb{R}^n)$ and $f \in C^\infty_p(\mathbb{R}^n)$.

$$v = \sum_{i=1}^{n} a^i(p)\frac{\partial}{\partial x^i}\big|_p \equiv X(p)$$

where $X : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)$ is formally

$$X(p) := \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p \in T_p(\mathbb{R}^n)$$

We then have a bilinear function

$$\langle , \rangle : T_p(\mathbb{R}^n) \times C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$$

given by

$$\langle v, f\rangle = \langle \sum_i a^i(p) \frac{\partial}{\partial x^i} | p, f\rangle := \sum_i a^i(p) \frac{\partial}{\partial x^i}\Big|_p f \in \mathbb{R}$$

One can then think of *tangent vectors* as $\langle v, ()\rangle$ functions of the 2nd argument. Likewise, we can think of df as maps on the 1st argument: $df_p() = \langle (), f\rangle$ that takes in tangent vectors and gives a number in $\mathbb{R}$.

Lemma B.3.2. *Let $x^1, x^2, \cdots, x^n : \mathbb{R}^n \to \mathbb{R}$ be the standard coordinate functions on $\mathbb{R}^n$. Then at each point p, $\{dx_p^1, dx_p^2 \cdots dx_p^n\}$ forms a basis for $\Lambda^1(T_p(\mathbb{R}^n))$ dual to $\{\frac{\partial}{\partial x^1}\big|_p, \frac{\partial}{\partial x^2}\big|_p, \cdots, \frac{\partial}{\partial x^n}\big|_p\} \in T_p(\mathbb{R}^n)$.*

Proof. Note that

$$dx^i : \mathbb{R}^n \to \cup_{p\in\mathbb{R}^n} \Lambda^1(T_p(\mathbb{R}^n))$$

where $dx_p^i \in \Lambda^1(T_p(\mathbb{R}^n))$ is such that

$$dx_p^i(\frac{\partial}{\partial x^j}\big|_p) := \frac{\partial}{\partial x^j}\Big|_p (x^i) = \delta_{ij}$$

and $dx_p^j \in \Lambda_p(T_p(\mathbb{R}^n))$ is precisely the dual basis to $\frac{\partial}{\partial x^i}\big|_p \in T_P(\mathbb{R}^n)$.

■

Note: If

$$\omega : U \to \amalg_{p\in U} \Lambda^1(T_p(\mathbb{R}^n))$$

is a $1-$form, $\omega(p) \in \Lambda^1(T_p(\mathbb{R}^n))$ and

$$\omega(p) = \sum_i a^i(p) dx_p^i$$

and formally

$$\omega(\cdot) \equiv \sum_i a_i(\cdot) dx_{(\cdot)}^i$$

Where $a^j(p) = w_p(\frac{\partial}{\partial x^j}\big|_p)$.

B.3.1 Differential Forms in terms of Coordinates

Theorem B.3.3. *Let $f : U \to \mathbb{R}$ be C^∞. Then df_p is defined as that linear function on $T_p(\mathbb{R}^n)$ such that*

$$df_p(\frac{\partial}{\partial x^i}\big|_p) = \frac{\partial}{\partial x^i}\Big|_p f$$

Then (as a covector)

$$df_p \equiv \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\Big|_p (dx^i)_p$$

and formally (as a map on U)

$$df_{()} \equiv \sum_{j=1}^{n} \frac{\partial f}{\partial x^i}\Big|_{()} dx_{()}^i$$

Proof. Let $f : U \to \mathbb{R}$. $df_p \in \Lambda^1(T_p(U))$ and hence we have an expansion of the form

$$df_p = \sum_{j=1}^{n} a_p^j dx_p^j \tag{B.3.1}$$

where $x^j : \mathbb{R}^n \to \mathbb{R}$ is the $j-$th standard coordinate function and dx^j is the dual covector to $\frac{\partial}{\partial x^i}\big|_p \in T_p(U)$. Applying df_p on $\frac{\partial}{\partial x^i}\big|_p$ (in B.3.1) we get

$$df_p(\frac{\partial}{\partial x^i}\big|_p) := \frac{\partial f}{\partial x^i}\big|_p = \sum_{j=1}^{n} a_p^j dx_p^j(\frac{\partial}{\partial x^i}\big|_p) = a_p^j$$

and hence

$$df_p \equiv \sum_{j=1}^{n} \frac{\partial f}{\partial x^j}\big|_p dx_p^j$$

■

Note: Let $X_p \in T_p(\mathbb{R}^n)$ be a tangent vector. we have $X_p \equiv \sum_{i=1}^n b^i(X_p)\frac{\partial}{\partial x^i}\big|_p$. Apply dx_p^j on both sides:

$$dx_p^j(X_p) = b^j(X_p)$$

and hence

$$X_p \equiv \sum_{i=1}^{n} dx_p^i(X_p)\frac{\partial}{\partial x^i}\big|_p$$

B.3.2 Differential $k-$forms

Definition B.3.4. (Differential $k-$form) A differential $k-$form is a function of the kind

$$\omega : U \to \amalg_{p\in U}\Lambda^k(T_p(\mathbb{R}^n))$$
$$\omega_p \in \Lambda^k(T_p(\mathbb{R}^n))$$

Since $\Lambda^k(T_p(\mathbb{R}^n))$ is spanned by vectors of the kind (for $1 \le i_1 < i_2 < \cdots < i_k \le n$)

$$dx_p^{i_1} \wedge dx_p^{i_2} \wedge \cdots \wedge dx_p^{i_k}$$

and hence

$$\omega_p \equiv \sum_{1\le i_1<\cdots<i_k\le n} w_{(p)}^{i_1 i_2\cdots i_k} dx_p^{i_1} \wedge dx_p^{i_2} \wedge \cdots \wedge dx_p^{i_k}$$

where $w^{i_1 i_2\cdots i_k} : U \to \mathbb{R}$ is a C^∞ function. Then we say ω is a C^∞ differential $k-$form.

Example B.3.5. (In $\mathbb{R}^3$) Let $x, y, z : \mathbb{R}^3 \to \mathbb{R}$ be the coordinate maps.

1. C^∞ $1-$forms are maps
$$\omega : U \to \amalg_{p\in U}\Lambda^1(T_p(U))$$
and
$$\omega_p \equiv \sum_{i=1}^{3} f_p^i dx_p^i = f_p^x dx_p + f_p^y dy_p + f_p^z dz_p \in \Lambda^1(T_p(U))$$

2. C^∞ $2-$forms are maps
$$\omega : U \to \amalg_{p\in U}\Lambda^2(T_p(U))$$
and
$$\omega_p \equiv \sum_{1\le i_1<i_2\le 3} f_p^{i_1 i_2} dx_p^{i_1} \wedge dx_p^{i_2} = f_p^{1,2} dx_p \wedge dy_p - f_p^{1,3} dz_p \wedge dx_p + f_p^{2,3} dy_p \wedge dz_p \in \Lambda^2(T_p(U))$$

3. C^∞ $3-$forms are maps
$$\omega : U \mapsto \amalg_p\Lambda^3(T_p(U))$$
and
$$\omega_p \equiv \sum_{1\le i_1<i_2<i_3} f_p^{i_1 i_2 i_3} dx_p^{i_1} \wedge dx_p^{i_2} \wedge dx_p^{i_3} = f_p dx_p \wedge dy_p \wedge dz_p \in \Lambda^3(T_p(U))$$

B.3.3 Exterior Derivative

Consider a C^∞ function $f : U \to \mathbb{R}$. We have $df : U \to \amalg_{p\in U}\Lambda^1(T_p(U))$ such that $df_p \in \Lambda^1(T_p(U))$.

$$df_p : T_p(U) \to \mathbb{R}$$

via

$$df_p(v) = df_p(\sum_{i=1}^{n} dx^i_p(v)\frac{\partial}{\partial x^i}\big|_p) = \sum_{i=1}^{n}\frac{\partial f}{\partial x^i}\big|_p dx^i_p(v)$$

and symbolically we have

$$df \equiv \sum_{i=1}^{n}\frac{\partial f}{\partial x^i}\big|dx^i$$

So d in this case can be viewed as a map that takes $0-$forms (functions) and gives us a $1-$form df.

Before we define the general exterior derivative for $k-$ forms, we will set up some notation.

Definition B.3.6. (Multi Index Notations) Let $I \equiv \{i_1, i_2, \cdots, i_k\} \in \mathscr{T}_k$ and $i \notin I$. For $i \in I$, we write $I_{\hat{i}}$ as the multi index I with i removed. For $i \notin I$, we write $[iI]$ as the multi index I with i added in the correct order.

We also write $\text{sign}[i, I]$ as the sign of the permutation that takes $\{i, I\}$ to $[iI]$. A corollary of this is that

$$dx^{[iI]} = \text{sign}[i, I]dx^i \wedge dx^I$$

We present an important lemma from [CDS18] that will be used in the document.

Lemma B.3.7. *Let $n \geq 2$ and $1 \leq k \leq n-1$ be integers. Then for any $I \in \mathscr{T}_{k+1}$ and $i, j \in I$, with $i \neq j$ we have*

$$\text{sign}[i, I_{\hat{i}\hat{j}}]\,\text{sign}[j, I_{\hat{i}\hat{j}}] = -\,\text{sign}[i, I_{\hat{i}}]\,\text{sign}[j, I_{\hat{j}}] \tag{B.3.2}$$

And and for any $I \in \mathscr{T}_{k-1}$ and every $i, j \notin I$ with $i \neq j$,

$$\text{sign}[i, [jI]]\,\text{sign}[j, [iI]] = -\,\text{sign}[i, I]\,\text{sign}[j, I] \tag{B.3.3}$$

Proof. We start with the first equation. Suppose that $i = i_r$ and $j = i_s$ for some r and s, such that (without loss of generality) $i < j$. Then we have $I = i_1 < i_2, \cdots, i_r = i, \cdots, i_s = j, i_{k+1}$. Then $\text{sign}[i, I_{\hat{i}}] = (-1)^{r-1}$ since to move $i = i_r$ across $r-1$ positions requires $r-1$ transpositions. Likewise $\text{sign}[j, I_{\hat{j}}] = (-1)^{s-1}$. Therefore the right hand side gives $(-1)^{r-1}(-1)^{s-1}$.

Since $I_{\hat{i}\hat{j}} = i_1 < i_2 < \cdots < i_{r-1} < i_{r+1} < \cdots < i_{s-1} < i_{s+1} < \cdots < i_{k+1}$,, $\text{sign}[i, I_{\hat{i}\hat{j}}] = (-1)^{r-1}$ but on the other hand $\text{sign}[j, I_{\hat{i}\hat{j}}] = (-1)^{s-2}$ (Since there is one less swap due to the absense of i). Therefore the left hand side reads $(-1)^{r-1}(-1)^{s-1}(-1)$ which gives the identity.

We now prove the second equation. Assume that $i < j$. It is clear that in that case, $\text{sign}[i, [jI]] = \text{sign}[i, I]$ (since we would not have to move i past j in $[i, [jI]]$). But in that case, $\text{sign}[j, [iI]]$ and $\text{sign}[j, I]$ would differ by one extra transposition, since j would have to be moved past i in $[j, [iI]]$. Therefore, $\text{sign}[j, [iI]] = -\,\text{sign}[j, I]$ and hence the identity follows.

■

Definition B.3.8. (The space of $k-$ forms) Let Ω be an open set in $\mathbb{R}^n$. The space of $k-$forms is denoted by

$$\{\Omega; \Lambda^k(T\Omega)\}$$

We also write the space of $k-$ forms as

$$\Omega(\Lambda^k)$$

Definition B.3.9. (Exterior derivative d) Let $\omega \in \{\Omega; \Lambda^k(T\Omega)\}$ be a $k-$form given by (see B.1.3)

$$\omega_p \equiv \sum_{I\in\mathscr{T}_k}\omega^I_p dx^I_p$$

$d : \{\Omega; \Lambda^k(T\Omega)\} \to \{\Omega; \Lambda^{k+1}\Omega\}$ is the function given by (formally)

$$d\omega_p := \big(\sum_{i=1}^{n} \frac{\partial}{\partial x^i}\big|_p dx_p^i\big) \wedge \omega_p \tag{B.3.4}$$

which is

$$\sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \frac{\partial \omega^I}{\partial x^i}\big|_p dx_p^i \wedge dx_p^I = \sum_{I \in \mathscr{T}_k} d\omega_p^I \wedge dx_p^I \tag{B.3.5}$$

Lemma B.3.10. *Let $\omega = \sum_{I \in \mathscr{T}_k} \omega^I dx^I$ be a $k-$form. Then*

$$d\omega = \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \frac{\partial \omega^I}{\partial x^i} dx^i \wedge dx^I = \sum_{I \in \mathscr{T}_{k+1}} \sum_{\gamma=1}^{k+1} (-1)^{\gamma-1} \frac{\partial \omega^{I_{\hat{i}_\gamma}}}{\partial x^{i_\gamma}} dx^I \tag{B.3.6}$$

Definition B.3.11. (Antiderivation) Let $A = \oplus_{k=0}^{\infty} A^k$ be a graded algebra over a field K. An antiderivation is a map $D : A \to A$ such that

1. D is $K-$linear
2. For all $a \in A^k$, $b \in A^l$,
$$D(ab) = (D(a))b + (-1)^k a(D(b))$$

D is said to be an antiderivation of degree m if $D|_{A^k} : A^k \to A^{k+m}$

Theorem B.3.12. *$d : \oplus_{k=0}^{\infty} \Omega(\Lambda^k) \to \oplus_{k=0}^{\infty} \Omega(\Lambda^k)$ is an antiderivation of degree 1 and*

$$d(\omega \wedge \tau) = d\omega \wedge \tau + (-1)^k \omega \wedge d\tau$$

for all $\omega \in \Omega(\Lambda^k)$ and $\tau \in \Omega(\Lambda^l)$

Proof. That d is linear is clear from formula B.3.5. Let $\omega \equiv \sum_{I \in \mathscr{T}_k} w^I dx^I$ and $\tau \equiv \sum_{J \in \mathscr{T}_l} \tau^J dx^J$. $\omega \wedge \tau \equiv \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} w^I \tau^J dx^I \wedge dx^J$ Applying d gives us

$$d(\omega \wedge \tau) = \big(\frac{\partial}{\partial x^i}\big|_p dx_p^i\big) \wedge \big(\sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} w_p^I \tau_p^J dx_p^I \wedge dx_p^J\big) =$$

$$\sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} \Big(\frac{\partial \omega^I}{\partial x^i}\big|_p \tau_p^J + \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p\Big) dx_p^i \wedge dx_p^I \wedge dx_p^J$$

$$= d\omega \wedge \tau + \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p dx_p^i \wedge dx_p^I \wedge dx_p^J$$

$$= d\omega \wedge \tau + \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \sum_{J \in \mathscr{T}_l} (-1)^k \omega_p^I \frac{\partial \tau^J}{\partial x^i}\big|_p dx_p^I \wedge dx_p^i \wedge dx_p^J$$

$$= d\omega \wedge \tau + (-1)^k \omega \wedge d\tau$$

■

Lemma B.3.13. $d^2 = 0$

Proof.

$$d\omega = \sum_{i=1}^{n} \sum_{I \in \mathscr{T}_k} \frac{\partial \omega^I}{\partial x^i} dx^i \wedge dx^I$$

and

$$d(d\omega) = \big(\sum_{i=1}^{n} \frac{\partial}{\partial x^i} dx^i\big) \wedge \big(\sum_{j=1}^{n} \sum_{I \in \mathscr{T}_k} \frac{\partial \omega^I}{\partial x^j} dx^j \wedge dx^I\big)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{I\in\mathscr{T}_k}\frac{\partial^2\omega^I}{\partial x^i x^j}dx^i\wedge dx^j\wedge dx^I$$

Since $dx^i\wedge dx^j = -dx^j\wedge dx^i$ and $\frac{\partial^2\omega^I}{\partial x^i x^j} = \frac{\partial^2\omega^I}{\partial x^j x^i}$ we have that $d(d\omega) = 0$.

■

B.3.4 Closed and Exact Forms

Definition B.3.14. (Closed Form) Let $\omega\in\Omega(\Lambda^k)$ be a $k-$form. We say ω is *closed* if $d\omega = 0$.

Definition B.3.15. (Exact Form) We say $k-$form $\omega\in\Omega(\Lambda^k)$ is *exact* if there exists $\lambda\in\Omega(\Lambda^{k-1})$ such that $d\lambda = \omega$.

Lemma B.3.16. Exact *forms are* closed.

B.4 Vector Calculus

We set $n = 3$, $U\subset\mathbb{R}^3$ and $k\in\{1,2,3\}$.

1. **One forms:** $\omega: U\to \amalg_{p\in U}T_p^*(\mathbb{R}^3)$ with $\omega_{(p)}\in\Lambda^1(T_p(\mathbb{R}^3))\equiv T_p^*(\mathbb{R}^3)$ given by
$$\omega_{(p)}\equiv\omega^x_{(p)}dx_{(p)}+\omega^y_{(p)}dy_{(p)}+\omega^z_{(p)}dz_{(p)}$$
2. **Two forms:** $\omega: U\to\amalg_{p\in U}\Lambda^2(T_p(\mathbb{R}^3))$ with $\omega_{(p)}\in\Lambda^2(T_p(\mathbb{R}^3))$ given by
$$\omega_{(p)}\equiv\sum_{1\le i_1<i_2\le 3}\omega^{i_1 i_2}_{(p)}dx^{i_1}_{(p)}\wedge dx^{i_2}_{(p)}$$
$$=\omega^z_{(p)}dx_{(p)}\wedge dy_{(p)}+\omega^x_{(p)}dy_{(p)}\wedge dz_{(p)}+\omega^y_{(p)}dz_{(p)}\wedge dx_{(p)}$$
3. **Three forms:** $\omega: U\to\amalg_{p\in U}\Lambda^3(T_p(\mathbb{R}^3))$ with $\omega_{(p)}\in\Lambda^3(T_p(\mathbb{R}^3))$ given by
$$\omega_{(p)} = w_{(p)}dx_{(p)}\wedge dy_{(p)}\wedge dz_{(p)}$$

We immediately see a correspondence in table B.1

Table B.1: Correspondence between vector fields and forms.

Field	Form
Vector field: $[\omega^x,\omega^y,\omega^z]$	$1-$form: $\omega^x_{(p)}dx_{(p)}+\omega^y_{(p)}dy_{(p)}+\omega^z_{(p)}dz_{(p)}$
Vector field: $[\omega^x,\omega^y,\omega^z]$	$2-$form: $\omega^x_{(p)}dy_{(p)}\wedge dz_{(p)}+\omega^y_{(p)}dz_{(p)}\wedge dx_{(p)}+\omega^z_{(p)}dx_{(p)}\wedge dy_{(p)}$
Scalar field: ω	$3-$form: $\omega_{(p)}dx_{(p)}\wedge dy_{(p)}\wedge dz_{(p)}$

B.4.1 Exterior Derivative of forms in $\mathbb{R}^3$

$0-$forms in $\mathbb{R}^3$

Let $f: U\to\mathbb{R}$. We know that
$$df: U\to\amalg_{p\in U}\Lambda^1(T_p(U))$$
given by
$$df_{(p)} = \frac{\partial f}{\partial x}\Big|_{(p)}dx_{(p)}+\frac{\partial}{\partial y}\Big|_{(p)}dy_{(p)}+\frac{\partial}{\partial z}\Big|_{(p)}dz_{(p)}$$
Which, from the correspondence B.1, can be identified with the vector field
$$\left[\frac{\partial f}{\partial x}\Big|_{(p)},\frac{\partial f}{\partial y}\Big|_{(p)},\frac{\partial f}{\partial z}\Big|_{(p)}\right]\equiv\nabla f(p)$$

1−forms in $\mathbb{R}^3$

Let $\omega \in \Omega(\Lambda^1(T\mathbb{R}^3))$ be a 1−form, i.e.

$$\omega_{(p)} \equiv \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$\text{(Symbolically) } d\omega_p = \Big(\sum_{i=1}^{3} \frac{\partial}{\partial x^i}\Big|_p dx^i_p \Big) \wedge \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$= \Big(\frac{\partial}{\partial x}\Big|_p dx_p + \frac{\partial}{\partial y}\Big|_p dy_p + \frac{\partial}{\partial z}\Big|_p dz_p \Big) \wedge \omega^x_{(p)} dx_{(p)} + \omega^y_{(p)} dy_{(p)} + \omega^z_{(p)} dz_{(p)}$$

$$= \Big[\frac{\partial \omega^y}{\partial x}\Big|_p dx_p \wedge dy_p + \frac{\partial \omega^z}{\partial x}\Big|_p dx_p \wedge dz_p \Big] + \Big[\frac{\partial \omega^x}{\partial y}\Big|_p dy_p \wedge dx_p + \frac{\partial \omega^z}{\partial y}\Big|_p dy_p \wedge dz_p \Big] + \Big[\frac{\partial \omega^x}{\partial z}\Big|_p dz_p \wedge dx_p + \frac{\partial \omega^y}{\partial z}\Big|_p dz_p \wedge dy_p \Big] =$$

$$\Big(\frac{\partial \omega^z}{\partial y}\Big|_p - \frac{\partial \omega^y}{\partial z}\Big|_p \Big) dy_p \wedge dz_p + \Big(\frac{\partial \omega^y}{\partial x}\Big|_p - \frac{\partial \omega^x}{\partial y}\Big|_p \Big) dx_p \wedge dy_p + \Big(\frac{\partial \omega^x}{\partial z}\Big|_p - \frac{\partial \omega^z}{\partial x}\Big|_p \Big) dz_p \wedge dx_p$$

which from the correspondence B.1 can be identified with the vector field

$$\Big[\Big(\frac{\partial \omega^z}{\partial y}\Big|_p - \frac{\partial \omega^y}{\partial z}\Big|_p \Big), \Big(\frac{\partial \omega^x}{\partial z}\Big|_p - \frac{\partial \omega^z}{\partial x}\Big|_p \Big), \Big(\frac{\partial \omega^y}{\partial x}\Big|_p - \frac{\partial \omega^x}{\partial y}\Big|_p \Big) \Big] \equiv \nabla \times \omega(p)$$

2−forms in $\mathbb{R}^3$

Let $\omega \in \Omega(\Lambda^2(T\mathbb{R}^3))$ be a 2−form i.e.

$$\omega_p \equiv \omega^x_p dy_p \wedge dz_p + \omega^y_p dz_p \wedge dx_p + \omega^z_p dx_p \wedge dy_p \in \Lambda^2(T_p(\mathbb{R}^3))$$

$$d(\omega^x dy \wedge dz)_p = \frac{\partial \omega^x}{\partial x}\Big|_p dx_p \wedge dy_p \wedge dz_p$$

$$d(\omega^y dz \wedge dx)_p = \frac{\partial \omega^y}{\partial y}\Big|_p dy_p \wedge dz_p \wedge dx_p$$

$$d(\omega^z dx \wedge dy)_p = \frac{\partial \omega^z}{\partial z}\Big|_p dz_p \wedge dx_p \wedge dy_p$$

and hence

$$d\omega_p \equiv \Big(\frac{\partial \omega^x}{\partial x}\Big|_p + \frac{\partial \omega^y}{\partial y}\Big|_p + \frac{\partial \omega^z}{\partial z}\Big|_p \Big) dx_p \wedge dy_p \wedge dz_p$$

Which from the correspondence B.1 can be identified with the scalar field

$$\frac{\partial \omega^x}{\partial x}\Big|_p + \frac{\partial \omega^y}{\partial y}\Big|_p + \frac{\partial \omega^z}{\partial z}\Big|_p \equiv \nabla \cdot \omega(p)$$

From these correspondences, we see that $d : \Omega(\Lambda^0) \to \Omega(\Lambda^1)$ is the gradient, $d : \Omega(\Lambda^1) \to \Omega(\Lambda^2)$ the curl and $d : \Omega(\Lambda^2) \to \Omega(\Lambda^3)$ is the divergence. From this we can see immediately that the well known identities

$$\nabla \times (\nabla f) = 0$$

and

$$\nabla \cdot (\nabla \times \omega) = 0$$

are just special cases of $d^2 = 0$.

B.5 Interior Product

We start with the observation that $\{dx^I : I \in \mathscr{T}_k\}$ not only forms a basis for $\Lambda^k(V)$, but is also an orthonormal set:

$$\langle dx^I, dx^J \rangle = \begin{cases} 1 & \text{if } I = J \\ 0 & \text{if } I \neq J \end{cases}$$

This is a natural consequence of the definition $\langle f, g \rangle = \sum_{I \in \mathscr{T}_k} f^I g^I$. In the case of a Riemannian manifold with a given $g_p : T_pM \times T_pM \to \mathbb{R}$, there is a natural choice for $\langle f, g \rangle_g$ given by

$$\langle f, g \rangle_g := \sum_{I,J \in \mathscr{T}_k} f^I g^J \det(g_p^{i_a, j_b}) \tag{B.5.1}$$

where $\{g^{pq}\}$ is the inverse of the matrix $\{g_{pq}\}$. We work predominantly in $\mathbb{R}^n$ with $g_{ij} = \delta_{ij}$ unless stated otherwise. The wedge product

$$\wedge : \Lambda^k(V) \times \Lambda^l(V) \to \Lambda^{k+l}(V)$$

is bilinear. Suppose we have a fixed k form $\alpha \in \Lambda^k(V)$, then

$$\alpha \wedge () : \Lambda^l(V) \to \Lambda^{k+l}(V)$$

is a linear map that *raises* the degree of the form. For a $k-$form α, we have the natural dual from the inner product given by:

$$\langle (\alpha \wedge)\omega, \beta \rangle = \langle \omega, (\alpha \lrcorner)\beta \rangle$$

where $\alpha \in \Lambda^k(V)$, $\omega \in \Lambda^{l-k}(V)$ and $\beta \in \Lambda^l$. From this definition it is clear that

$$\alpha \lrcorner : \Lambda^l(V) \to \Lambda^{l-k}(V)$$

Corollary B.5.1. If $k \geq l$ and $\alpha \in \Lambda^k(V)$ and $\beta \in \Lambda^l(V)$, then $\alpha \lrcorner \beta = 0$.

On *basis elements*, we have for $dx^I \in \Lambda^k(V)$, $dx^J \in \Lambda^l(V)$ and $dx^w \in \Lambda^{l-k}(V)$

$$\langle dx^I \wedge dx^w, dx^J \rangle = \langle dx^w, dx^I \lrcorner dx^J \rangle$$

Proposition B.5.2. For $dx^I \in \Lambda^k(V)$, $dx^J \in \Lambda^l(V)$ and $l > k$, we have

$$dx^I \lrcorner dx^J = \begin{cases} \text{sign}(R(I, J/I)) dx^{J/I} & \text{if } I \subseteq J \\ 0 & \text{if } I \not\subseteq J \end{cases}$$

where J/I denotes the ordered set of indices that are in J but not in I and $R(I, J/I)$ denotes the permutation that reorders the tuple $(I, J/I)$.

Proof. **When $\mathbf{I \not\subseteq J}$**, we have some index of dx^I not be in dx^J. Therefore, no matter our choice of dx^w, we must have that

$$\langle dx^I \wedge dx^w, dx^J \rangle = 0 = \langle dx^w, dx^I \lrcorner dx^J \rangle$$

which means that $dx^I \lrcorner dx^J = 0$.

When $\mathbf{I \subseteq J}$, If dx^w does not contain all the complementary elements of J/I, we must have that $\langle dx^I \wedge dx^w, dx^J \rangle = \langle dx^w, dx^I \lrcorner dx^J \rangle = 0$. This means that $dx^I \lrcorner dx^J$ contains indices exclusively from J/I (upto reordering). Let $dx^I \lrcorner dx^J = \lambda dx^{J/I}$. Choose $dx^w = dx^{J/I}$. Note in that case that $dx^I \wedge dx^w = \text{sgn}(R(I, J/I)) dx^J$ where $R(I, J/I)$ is the permutation that rearranges $(I, J/I)$ to the canonically ordered tuple J.

We therefore have

$$\langle dx^I \wedge dx^w, dx^J \rangle = \text{sgn}(R(I, J/I)) \langle dx^J, dx^J \rangle = \text{sgn}(R(I, J/I))$$

$$= \langle dx^{J/I}, \lambda dx^{J/I} \rangle = \lambda$$

This implies $\lambda = \text{sgn}(R(I, J/I))$ and the proof is complete.

■

Let $\alpha \in \Lambda^k(V)$, $\beta \in \Lambda^l(V)$. For $\omega \in \Lambda^m$ $L_{\alpha\wedge\beta}(\omega) = \alpha \wedge (\beta \wedge \omega) = L_\alpha \circ L_\beta(\omega)$. Therefore, the Adjoint must follow

$$L^\dagger_{\alpha\wedge\beta} \equiv (L_\alpha \circ L_\beta)^\dagger = L^\dagger_\beta \circ L^\dagger_\alpha$$

which implies

$$(\alpha \wedge \beta) \lrcorner \gamma = \beta \lrcorner (\alpha \lrcorner \gamma)$$

B.5.1 The Musical Isomorphism and Interior Product

The situation in $\mathbb{R}^n$:

Given a C^∞ vector field $X : U \to \amalg_{p\in U} T_p(U)$ given by

$$X_{(p)} = \sum_{i=1}^n a^i(p) \frac{\partial}{\partial x^i}|_p \in T_p(U)$$

we can find a differential $1-form$ given by

$$p \mapsto \sum_{i=1}^n a^i(p) dx^i_{(p)} \tag{B.5.2}$$

The $\flat$ map

For any general Manifold, the vector spaces T_pM and T^*_pM are merely of the same dimension, but there is no global way to identify T_pM with T^*_pM. A metric changes this.

Let $g_p : T_pM \times T_pM \to \mathbb{R}$ be an inner product on T_pM. For a given vector $X \in T_pM$, define $X^\flat \in \Lambda^1(T_p(M))$ given by

$$X^\flat(Y) = g(X, Y) \tag{B.5.3}$$

for $Y \in T_p(M)$. Therefore, $\flat : T_p(M) \to T^*_p(M)$. This map is linear, and injective since $X^\beta(Y) = g(X, Y) = 0$, for every $Y \in T_p(M)$ implies $X = 0$. Hence, $\flat : T_pM \to T^*_pM$ is an isomorphism.

Let $X \equiv \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}$ be a vector field. g being an inner product, is a $2-$tensor on $T_p(M)$.

By B.1.4 we see then, that

$$g \equiv \sum_{1\le i,j\le n} g_{ij}(p) dx^i_p \otimes dx^j_p : T_pM \times T_pM \to \mathbb{R}$$

and B.5.3 tells us that

$$X^\flat_p = \sum_{1\le i,j\le n} g_{ij}(p) \sum_{s=1}^n a^s(p) dx^i_p (\frac{\partial}{\partial x^s}|_p) dx^j_p = \sum_{1\le i\le n} \sum_{1\le j\le n} g_{ij}(p) a^i(p) dx^j_p \in \Lambda^1(T_p(M))$$

If we assume that $g_{ij} = \delta_{ij}$ we get back B.5.2.

On Basis: If $X \equiv \frac{\partial}{\partial x^q}$, then $X^\flat = g(X, ())$ given by

$$\sum_{i=1}^n \sum_{j=1}^n g_{ij} dx^i (\frac{\partial}{\partial x^q}) dx^j = \sum_{j=1}^n g_{qj} dx^j$$

Therefore,

$$\Big(\frac{\partial}{\partial x^i}\Big)^\flat \equiv \sum_{j=1}^n g_{ij} dx^j$$

If our metric g were Euclidean ($g_{ij} = \delta_{ij}$) we would have that

$$\Big(\frac{\partial}{\partial x^i}\Big)^\flat \equiv dx^i$$

The $\sharp$ map

We know that $\flat : T_pM \to T_p^*M$ is given by $\left(\frac{\partial}{\partial x^i}|_p\right)^\flat = \sum_{j=1}^n g_{ij} dx_p^j$, so we have the matrix representation of $\flat$ as

$$[X^\flat]_{\{dx_p^1, dx_p^2, \cdots, dx_p^n\}} = \begin{bmatrix} g_{11} & g_{21} & \cdots & g_{n1} \\ g_{12} & g_{22} & \cdots & g_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ g_{1n} & g_{2n} & \cdots & g_{nn} \end{bmatrix} [X]_{\{\partial_{x^1}, \partial_{x^2}, \cdots, \partial_{x^n}\}} = G_p [X]_{\{\partial_{x^1}, \partial_{x^2}, \cdots, \partial_{x^n}\}}$$

from which we get the inverse map $\sharp : T_p^*M \to T_pM$ by

$$(dx_p^i)^\sharp = \sum_{j=1}^n (G_p)_{ij}^{-1} \frac{\partial}{\partial x^j}\bigg|_p \in T_pM$$

Interior product of a vector and a $k-$form (in $\mathbb{R}^n$)

Let $X \equiv \frac{\partial}{\partial x^q}$. In $\mathbb{R}^n$ we have $X^\flat = dx^q$. For a basis $k-$form $dx^{i_1} \wedge dx^{i_2} \cdots \wedge dx^{i_k}$,

$$\begin{aligned} X^\flat \lrcorner (dx^{i_1} \wedge dx^{i_2} \cdots \wedge dx^{i_k}) &= dx^q \lrcorner (dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k}) \\ &= \begin{cases} (-1)^{r-1} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge \widehat{dx^{i_r}} \wedge \cdots \wedge dx^{i_k} & \text{whenever } q = i_r \\ 0 & \text{whenever } q \notin \{i_1, i_2, \cdots, i_k\} \end{cases} \end{aligned} \tag{B.5.4}$$

Definition B.5.3. (Interior product of a vector with a form ($\imath_X(\omega)$)) Let $\omega \in \Omega(\Lambda^k)$ be a $k-$form. Let X be a vector field. $\imath_X : \Omega(\Lambda^k) \to \Omega(\Lambda^{k-1})$ is given by

$$\imath_X(\omega)(X_1, X_2, \cdots, X_{k-1}) = \omega(X, X_1, X_2, \cdots, X_{k-1})$$

Remark B.5.4. We see that $\imath_X(\omega)$ is a $k-1$ form. Hence, its action on basis vectors $\frac{\partial}{\partial x^i}$ tells us its action on arbitrary vectors by linearity. Moreover, $\imath_{X+cY}(\omega) = \imath_X(\omega) + c\imath_Y(\omega)$. So its action on $X = \frac{\partial}{\partial x^q}$ tells us its action on arbitrary X by linearity.

Theorem B.5.5. *In $\mathbb{R}^n$ equipped with the euclidean metric, we have*

$$\imath_X(\omega) = X^\flat \lrcorner \omega$$

(B.5.4 and B.5.3 are equivalent.)

Proof. By B.5.4 it is enough to check equality on basis vectors. Let $X \equiv \frac{\partial}{\partial x^q}$ and let $\omega = dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k}$ be a $k-$form. Let $X_1 = \frac{\partial}{\partial x^{j_1}}, X_2 = \frac{\partial}{\partial x^{j_2}}, \cdots, X_{k-1} = \frac{\partial}{\partial x^{j_{k-1}}}$ be $k-1$ subcollection of basis vectors. From B.5.4 we have

$$\begin{aligned} &X^\flat \lrcorner (dx^{i_1} \wedge dx^{i_2} \cdots \wedge dx^{i_k}) \\ &= \begin{cases} (-1)^{r-1} dx^{i_1} \wedge \cdots \widehat{dx^{i_r}} \cdots \wedge dx^{i_k} & \text{whenever } q = i_r \\ 0 & \text{whenever } q \notin \{i_1, i_2, \cdots, i_k\} \end{cases} \end{aligned}$$

So when evaluated on $\frac{\partial}{\partial x^{j_1}}, \frac{\partial}{\partial x^{j_2}}, \cdots, \frac{\partial}{\partial x^{j_{k-1}}}$ we have

$$X^\flat \lrcorner (dx^{i_1} \wedge \cdots \wedge dx^{i_k})(\partial_{x^{j_1}}, \partial_{x^{j_2}}, \cdots, \partial_{x^{j_{k-1}}}) =$$

$$= \begin{cases} (-1)^{r-1} \det(\{dx^{i_s}(\partial_{x^{j_t}}) : s \neq r, 1 \leq t \leq k-1\}) & \text{whenever } q = i_r \\ 0 & \text{whenever } q \notin \{i_1, i_2, \cdots, i_k\} \end{cases}$$

Let us look at $\imath_{\partial_{x^q}}(dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k})(\partial_{x^{j_1}}, \partial_{x^{j_2}}, \cdots, \partial_{x^{j_{k-1}}}) = \det(D)$ where D is the matrix

$$\begin{bmatrix} dx^{i_1}(\partial_{x^q}) & dx^{i_1}(\partial_{x^{j_1}}) & \cdots & dx^{i_1}(\partial_{x^{j_{k-1}}}) \\ dx^{i_2}(\partial_{x^q}) & dx^{i_2}(\partial_{x^{j_1}}) & \cdots & dx^{i_2}(\partial_{x^{j_{k-1}}}) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{i_r}(\partial_{x^q}) & dx^{i_r}(\partial_{x^{j_1}}) & \cdots & dx^{i_r}(\partial_{x^{j_{k-1}}}) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{i_k}(\partial_{x^q}) & dx^{i_k}(\partial_{x^{j_1}}) & \cdots & dx^{i_k}(\partial_{x^{j_{k-1}}}) \end{bmatrix}$$

Assuming that $q = i_r$ we see that D takes the form:

$$\begin{bmatrix} 0 & dx^{i_1}(\partial_{x^{j_1}}) & \cdots & dx^{i_1}(\partial_{x^{j_{k-1}}}) \\ 0 & dx^{i_2}(\partial_{x^{j_1}}) & \cdots & dx^{i_2}(\partial_{x^{j_{k-1}}}) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & dx^{i_k}(\partial_{x^{j_1}}) & \cdots & dx^{i_k}(\partial_{x^{j_{k-1}}}) \end{bmatrix}$$

Whence we see that the determinant takes the following form:

$$(-1)^{r-1} \det(\{dx^{i_s}(\partial_{x^{j_t}}) : s \neq r, 1 \leq t \leq k-1\})$$

which agrees with $X^\flat \lrcorner (dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k})$.

The case $q \notin \{i_1, i_2, \cdots, i_k\}$ follows since $q \neq i_r$ implies that the first column in D full of 0, which makes the determinant 0.

■

Lemma B.5.6. ***(Expression for interior product of a*** $1-$***form and a*** $k-$***form)*** *Let* $\nu = \sum_{j=1}^n v^j dx^j \in \Lambda^1$ *and* $\omega \in \Lambda^k$*. We have*

$$\nu \lrcorner \omega = \sum_{j=1}^n \sum_{I \in \mathscr{T}_k} v^j \omega^I dx^j \lrcorner dx^I = \sum_{I \in \mathscr{T}_k} \sum_{j \in I} v^j \omega^I (\text{sign}[j, I_{\hat{j}}]) dx^{I_{\hat{j}}} = \sum_{j=1}^n \sum_{I \in \mathscr{T}_{k-1}} v^j \omega^{j i_1 i_2 \cdots i_{k-1}} dx^I \tag{B.5.5}$$

Caution: *The notation used here is adopted from [CDS18] and [CDK12] wherein, for non-increasing indices, we define*

$$\omega^I = \text{sign}([I]) \omega^{[I]} \tag{B.5.6}$$

where I *is an arbitrary multi-index set and* $[I]$ *is (by abuse of notation) both the permutation that rearranges* I *to be increasing and the increasing index set.*

Proof.

$$\nu \lrcorner \omega = \sum_{j=1}^n \sum_{I \in \mathscr{T}_k} v^j \omega^I dx^j \lrcorner dx^I = \sum_{I \in \mathscr{T}_k} \sum_{j \in I} v^j \omega^I (\text{sign}[j, I_{\hat{j}}]) dx^{I_{\hat{j}}} \tag{B.5.7}$$

Note that, for every $K \in \mathscr{T}_{k-1}$ and $j \notin K$, we have a unique $I \in \mathscr{T}_k$ given by $[jK]$ so that $K = I_{\hat{j}}$. We now argue that the expression above is equal to the following:

$$\sum_{j=1}^n \sum_{K \in \mathscr{T}_{k-1}} v^j \omega^{[jK]} \text{sign}[j, K] dx^K$$

For a given $j \in \{1, \cdots, n\}$, $dx^j \lrcorner dx^I \neq 0$ only for those $I \in \mathscr{T}_k$ with $j \in I$. And for each such I, there is a unique $K \in \mathscr{T}_{k-1}$ such that $I = [jK]$. Therefore we can write the left hand side of B.5.7 as

$$\sum_{j=1}^n \sum_{K \in \mathscr{T}_{k-1}} v^j \omega^{[jK]} dx^j \lrcorner dx^{[jK]} = \sum_{j=1}^n \sum_{K \in \mathscr{T}_{k-1}} v^j \omega^{[jK]} \text{sign}[j, K] dx^K$$

If we further impose the notation in (B.5.6), we get

$$\sum_{j=1}^n \sum_{K \in \mathscr{T}_{k-1}} v^j \omega^{j,K} dx^K$$

which are the desired results.

■

Theorem B.5.7. *Let X be a vector field, $\omega \in \Omega\Lambda^k$ and $\eta \in \Omega\Lambda^l$. We then have*

$$\imath_X(\omega \wedge \eta) = \imath_X(\omega) \wedge \eta + (-1)^k \omega \wedge \imath_X(\eta) \tag{B.5.8}$$

This tells us that $\imath_X : \oplus_{k=0}^{\infty}\Omega(\Lambda^k) \to \oplus_{k=0}^{\infty}\Omega(\Lambda^k)$ is a rank -1 antiderivation.

Proof. Let $X_{(p)} = \sum_{q=1}^n a^q(p)\frac{\partial}{\partial x^q}\big|_p$, $\omega = \sum_{I\in\mathscr{I}_k} \omega^I dx^I$ and $\eta = \sum_{J\in\mathscr{I}_l} \eta^J dx^J$.
$\imath_X(\omega\wedge\eta) = \sum_{q=1}^n a^q \imath_{\partial_{x^q}}(\omega\wedge\eta) = \sum_{q=1}^n a^q \imath_{\partial_{x^q}}(\sum_{I\in\mathscr{I}_k}\sum_{J\in\mathscr{I}_l}\omega^I\eta^J dx^I\wedge dx^J)$ which gives

$$\imath_X(\omega\wedge\eta) = \sum_{q=1}^n \sum_{I\in\mathscr{I}_k}\sum_{J\in\mathscr{I}_l} a^q\omega^I\eta^J \imath_{\partial_{x^q}}(dx^I\wedge dx^J)$$

If we suppose that the identity holds on basis elements i.e. $X = \partial_{x^q}$ $\omega = dx^I$ and $\eta = dx^J$, we then have

$$\imath_X(\omega\wedge\eta) = \sum_{q=1}^n \sum_{I\in\mathscr{I}_k}\sum_{J\in\mathscr{I}_l} a^q\omega^I\eta^J\Big(\imath_{\partial_{x^q}}(dx^I)\wedge dx^J + (-1)^k dx^I \imath_{\partial_{x^q}}(dx^J)\Big)$$

Which gives the result. Hence we need only to prove the identity for $X = \partial_{x^q}$, $\omega = dx^I$ and $\eta = dx^J$.

Case 1: $\mathbf{q \in I}$ and $\mathbf{I \cap J = \varnothing}$
Suppose that $q = i_r \in I$. We have $i_{\partial_{x^q}}(dx^I \wedge dx^J) = \det D$ where

$$D = \begin{bmatrix} dx^{i_1}(\partial_{x^q}) & dx^{i_1}(x_1) & \cdots & dx^{i_1}(x_{k+l-1}) \\ dx^{i_2}(\partial_{x^q}) & dx^{i_2}(x_1) & \cdots & dx^{i_2}(x_{k+l-1}) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{i_r}(\partial_{x^q}) & dx^{i_r}(x_1) & \cdots & dx^{i_r}(x_{k+l-1}) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{j_1}(\partial_{x^q}) & dx^{j_1}(x_1) & \cdots & dx^{j_1}(x_{k+l-1}) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{j_l}(\partial_{x^q}) & dx^{j_l}(x_1) & \cdots & dx^{j_l}(x_{k+l-1}) \end{bmatrix}$$

which evaluates to (from Laplace's Formula)

$$(-1)^{r-1} dx^{i_1}\wedge dx^{i_2}\wedge\cdots\widehat{dx^{i_r}}\wedge\cdots\wedge dx^{i_k}\wedge dx^{j_1}\wedge\cdots\wedge dx^{j_l}(x_1,x_2,\cdots,x_{k+l-1}) = \imath_{\partial_{x^q}}(dx^I)\wedge dx^J$$

If $J\cap I = \varnothing$ and $q = i_r \in I$, note that $dx^I \wedge i_{\partial_{x^q}}(dx^J) = dx^I \wedge (dx^q \lrcorner (dx^J)) = 0$ since q is not in J (see Formula B.5.1). Therefore, we have the identity in this case.

Case 2: $\mathbf{q \in J}$ and $\mathbf{I \cap J = \varnothing}$
Let $q = j_s$ for some $j_s \in J$. We have $\imath_{\partial_{x^q}}(dx^I\wedge dx^J) = \det D$ where D is as in case 1, but note that the only non zero element in the first column occurs in the $k+s$ row. The Laplace expansion thus reads

$$(-1)^{k+s-1} dx^{i_1}\wedge dx^{i_2}\wedge\cdots\wedge dx^{i_k}\wedge dx^{j_1}\wedge dx^{j_2}\cdots\wedge\widehat{dx^{j_s}}\wedge\cdots\wedge dx^{j_l}(x_1,x_2,\cdots,x_{k+l-1}) = (-1)^k dx^I\wedge\imath_{\partial_{x^q}}(dx^J)$$

If $I\cap J = \varnothing$ and $q = j_s$ for some $j_s \in J$ then note that $\imath_{\partial_{x^q}}(dx^I)\wedge dx^J = (dx^q \lrcorner dx^I)\wedge dx^J = 0$ and likewise we have the identity.

Note that in both case 1 and case 2 we only used the following fact: $q \in I$ (or J) and $q \notin J$ (or $\notin I$).

Case 3: $\mathbf{I\cap J \neq \varnothing}$ and $\mathbf{q = i_r = j_s}$
Note that the left hand side $\imath_{\partial_{x^q}}(dx^I\wedge dx^J) = 0$. The first expression on the right hand side reads $\imath_{\partial_{x^q}}(dx^I)\wedge dx^J = (dx^q\lrcorner dx^I)\wedge dx^J$ and the second expression reads $(-1)^k dx^I \wedge (dx^q \lrcorner dx^J)$. Note that in case 3, the matrix D we get has two occurences of 1 (and the rest 0) in the first column: In the $r-$th and $(k+s)-$th rows. Laplace's expansion gives us the determinant of D to be

$$(-1)^{r-1} dx^{i_1}\wedge\cdots\wedge\widehat{dx^{i_r}}\wedge\cdots\wedge dx^{i_k}\wedge dx^{j_1}\wedge\cdots\wedge dx^{j_l}(x_1,x_2,\cdots,x_{k+l-1})$$

$$+(-1)^{k+s-1}dx^{i_1}\wedge dx^{i_2}\wedge dx^{i_k}\wedge dx^{j_1}\wedge\cdots\widehat{dx^{j_s}}\wedge\cdots\wedge dx^{j_l}(x_1,x_2,\cdots,x_{k+l-1})$$

which is precisely

$$\imath_{\partial_{x^q}}(dx^I)\wedge dx^J+(-1)^k dx^I\wedge(\imath_{\partial_{x^q}}dx^J)$$

Since the identity is true on basis, it is true in the general case through multilinearity.

■

Lemma B.5.8. *$\imath$ is anticummutative, i.e.*

$$\imath_X\imath_Y=-\imath_Y\imath_X$$

Proof. This is easy to see since $\imath_X(\omega)(x_1,\cdots,x_{k-1})=\omega(X,x_1,\cdots,x_{k-1})$ and $\imath_Y\imath_X(\omega)(x_1,\cdots,x_{k-2})=\imath_X(\omega)(Y,x_1,x_2,\cdots,x_{k-2})=\omega(X,Y,x_1,\cdots,x_{k-2})=-\omega(Y,X,x_1,\cdots,x_{k-2})$ and the proof is complete.

■

Corollary B.5.9. $\imath_X\circ\imath_X=0$

Theorem B.5.10. *Let $\xi\in\Omega(\Lambda^1)$, $\eta\in\Omega(\Lambda^k)$ be 1 and k forms respectively in $\mathbb{R}^n$ (with the Euclidean Metric). Then we have the following identity:*

$$||\xi||^2\eta=\xi\wedge(\xi\lrcorner\eta)+\xi\lrcorner(\xi\wedge\eta) \tag{B.5.9}$$

For vector fields we have:

$$|X|^2\eta=X^\flat\wedge\imath_X(\eta)+\imath_X(X^\flat\wedge\eta)$$

Proof. Using the identity B.5.7 we have $\xi\lrcorner(\xi\wedge\eta)=(\xi\lrcorner\xi)\wedge\eta+(-1)\xi\wedge(\xi\lrcorner\eta)$ which gives

$$\xi\lrcorner(\xi\wedge\eta)+\xi\wedge(\xi\lrcorner\eta)=(\xi\lrcorner\xi)\wedge\eta=\lambda\eta$$

Note that $\xi\wedge 1=\xi$ and $\langle\xi\wedge 1,\beta\rangle=\langle 1,\xi\lrcorner\beta\rangle$. Choosing $\beta=\xi$ gives

$$\langle 1,\xi\lrcorner\xi\rangle=\xi\lrcorner\xi=\langle\xi,\xi\rangle=||\xi||^2$$

So we have the identity.

■

B.6 Hodge Star

Let f be a $k-$form given by

$$f\equiv\sum_{I\in\mathscr{T}_k}f^I dx^I$$

We define the operation *Hodge Star* as follows (defined first on basic $k-$forms, and then extended linearly for any $k-$form):

$$\star\, dx^{i_1}\wedge dx^{i_2}\wedge\cdots\wedge dx^{i_k}=\text{sign}(R(I,J))dx^{j_1}\wedge dx^{j_2}\wedge\cdots\wedge dx^{j_{n-k}}$$

where

1. $I\equiv\{i_1,i_2,\cdots,i_k\}\in\mathscr{T}_k$ and $J\equiv\{j_1,j_2,\cdots,j_{n-k}\}$ being the complementary increasing index set to I (i.e. $J\cup I\equiv\{1,2,\cdots,n\}$ where J is also increasingly ordered).
2. $R(I,J)\in sym(n)$ rearranges $\{i_1,i_2,\cdots,i_k,j_1,j_2,\cdots,j_{n-k}\}$ in its increasing order.
3. For any $f\equiv\sum_{I\in\mathscr{T}_k}f^I dx^I$ we have
$$\star\, f:=\sum_{I\in\mathscr{T}_k}f^I\star dx^I$$

An alternate way to define the *Hodge Star*: If f and g are $k-$forms, then $\star\, g$ is defined as the unique $(n-k)-$form such that

$$f \wedge \star g = \langle f, g\rangle dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n \tag{B.6.1}$$

where the inner product is

$$\langle f, g\rangle := \sum_{I \in \mathscr{I}_k} f^I g^I$$

Lemma B.6.1. *Let $f \in \Omega(\Lambda^k)$. We then have*

$$\star\star f = (-1)^{k(n-k)} f \tag{B.6.2}$$

Proof. Since $\star$ is linear, it is enough to prove on basis elements.

$$\star dx^I = \text{sign}(R(I,J)) dx^J$$

where J is the complement of I relative to $\{1, 2, \cdots, n\}$ and $R(I,J)$ is the permutation that rearranges $\{i_1, i_2, \cdots, i_k, j_1, j_2, \cdots, j_{n-k}\} \to \{1, 2, \cdots, n\}$.

$$\star\star\, dx^I = \text{sign}(R(I,J)) \star dx^J = \text{sign}(R(I,J))\,\text{sign}(R(J,I)) dx^I$$

To compute the product of the signs we look at the map σ that takes (I,J), rearranges them to $\{1, 2, \cdots, n\}$ and then rearranges them to $\{J, I\}$. This amounts to moving the first k elements behind the last $n-k$ elements. This takes $k(n-k)$ transpositions, and hence the sign is $(-1)^{k(n-k)}$. Therefore,

$$\star\star\, dx^I = (-1)^{k(n-k)} dx^I$$

and the result follows for a general $k-$form from linearity.

■

Lemma B.6.2. *For $f, g \in \Omega(\Lambda^k)$*

$$\langle f, g\rangle = \langle \star f, \star g\rangle$$

Proof. Since $f \wedge \star g = \langle f, g\rangle dV$ (where dV is the volume form, in the Euclidean case, $dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n$), we have $\star f \wedge \star\star\, g = (-1)^{k(n-k)} \star f \wedge g = \langle \star f, \star g\rangle dV$. Recall that $f \wedge g = (-1)^{kl} g \wedge f$ for $f \in \Omega\Lambda^k$ and $g \in \Omega\Lambda^l$. Since $\star f \in \Lambda^{n-k}$ and $g \in \Lambda^k$, we have $\star f \wedge g = (-1)^{k(n-k)} g \wedge \star f$. This gives us

$$(-1)^{k(n-k)} \star f \wedge g = g \wedge \star f = \langle \star f, \star g\rangle dV$$

But since $g \wedge \star f := \langle g, f\rangle = \langle f, g\rangle$ (Real inner product), we have the result.

■

B.6.1 Hodge Star and Interior

Lemma B.6.3. *Let ω be a $k-$form and η an $l-$form. We have the following identity:*

$$\omega \lrcorner \eta = (-1)^{(n-l)(l-k)} \star (\omega \wedge \star\eta) \tag{B.6.3}$$

If ω were a 1 form, this formula reads:

$$\omega \lrcorner \eta = (-1)^{n(l-1)} \star (\omega \wedge \star\eta) \tag{B.6.4}$$

Proof. Let γ be an arbitrary $(l-k)-$form.

$$\langle \omega \wedge \gamma, \eta\rangle dV = \omega \wedge \gamma \wedge \star\eta$$

By the anti-commutation rule of the wedge, we have

$$\langle \omega \wedge \gamma, \eta\rangle dV = (-1)^{k(l-k)} \gamma \wedge (\omega \wedge \star\eta)$$

Using the $\star\star$ identity, we can write

$$(\omega \wedge \star\eta) = (-1)^{(k+n-l)(l-k)} \star\star(\omega \wedge \star\eta)$$

which gives us

$$\langle \omega \wedge \gamma, \eta\rangle dV = (-1)^{k(l-k)}(-1)^{(k+n-l)(l-k)}\gamma \wedge \star\big(\star(\omega \wedge \star\eta)\big)$$

$\star(\omega \wedge \star\eta)$ is a $(l-k)$ form and we can invoke the hodge star definition to get:

$$\langle \omega \wedge \gamma, \eta\rangle dV = (-1)^{k(l-k)}(-1)^{(k+n-l)(l-k)}\langle \gamma, \star(\omega \wedge \star\eta)\rangle dV$$

This gives us

$$\langle \gamma, \omega \lrcorner \eta\rangle = (-1)^{(l-k)(n-l)}\langle \gamma, \star(\omega \wedge \star\eta)\rangle$$

Since γ was arbitrary, we have

$$\omega \lrcorner \eta = (-1)^{(l-k)(n-l)} \star (\omega \wedge \star\eta)$$

■

Corollary B.6.4. Let X be a vector field and ω a $k-$form.

$$\imath_X(\omega) = (-1)^{(n-k)(k-1)} \star (X^\flat \wedge \star\omega) \equiv (-1)^{n(l-1)} \star (X^\flat \wedge \star\omega)$$

B.7 Brief Look at Riemannian Manifolds

Definition B.7.1. (Riemannian Manifold) A Smooth $n-$dimensional manifold $(M, \mathscr{A})$ is said to be a *Riemannian Manifold* of dimension n if for every $p \in M$ there exists an inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ (smoothly depending on p) that is positive definite. Such a g is called the *metric*.

We recall the fact (B.1.4) that the space of covariant–2 tensors has a basis given by $\{dx^i \otimes dx^j : i, j \leq n\}$. This then lets us write the metric in the following useful form:

$$g_p = \sum_{i,j=1}^{n} g_{ij} dx^i \otimes dx^j \tag{B.7.1}$$

Where $g_{ij} = g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j})$.

We also recall the musical isomorphisms (B.5.1) here:

The $\flat$ map

Let X be a vector field on M. Define the $\flat$ map that takes vector fields to $1-$forms as follows:

$$X^\flat := g(X, \cdot)$$

Let $X \equiv \frac{\partial}{\partial x^q}$, then

$$(\frac{\partial}{\partial x^q})^\flat = \sum_{i\leq n}\sum_{j\leq n} g_{ij} dx^i (\frac{\partial}{\partial x^q}) dx^j = \sum_{j\leq n} g_{qj} dx^j \tag{B.7.2}$$

And for $\omega = \sum_{q=1}^{n} \omega^q \frac{\partial}{\partial x^q}$ we have

$$(\omega)^\flat = \sum_{j\leq n} (\sum_{q=1}^{n} g_{qj}\omega^q) dx^j \tag{B.7.3}$$

The $\sharp$ map

Since g was positive–definite, we have that the matrix g_{ij} is invertible. Hence, we have the inverse of the $\flat$ map, called $\sharp$ map that takes $1-$forms to vector fields as follows: Let $\alpha = \sum_{i=1}^{n} \alpha^i dx^i$. Then

$$\alpha^\sharp = \sum_{j=1}^{n} (\sum_{q=1}^{n} g^{qj}\alpha^q) \frac{\partial}{\partial x^j} \tag{B.7.4}$$

Where $g^{ij} = ([g_{ij}]^{-1})_{ij}$

B.7.1 Natural Inner Product on Forms, Hodge Star and Interior Product

Let M be an n dimensional Riemannian Manifold with metric g. Let $\alpha = v_1 \wedge v_2 \cdots \wedge v_k$ be a k form generated by wedging $1-$forms $v_1, v_2, \cdots, v_k$. Likewise let $\beta = w_1 \wedge w_2 \wedge \cdots \wedge w_k$. We define the inner product on $k-$forms as follows:

$$\langle \alpha, \beta \rangle_g := \det \left(\begin{bmatrix} g(v_1^\sharp, w_1^\sharp) & g(v_1^\sharp, w_2^\sharp) & \cdots & g(v_1^\sharp, w_k^\sharp) \\ g(v_2^\sharp, w_1^\sharp) & g(v_2^\sharp, w_2^\sharp) & \cdots & g(v_2^\sharp, w_k^\sharp) \\ \vdots & \vdots & \ddots & \vdots \\ g(v_k^\sharp, w_1^\sharp) & g(v_k^\sharp, w_2^\sharp) & \cdots & g(v_k^\sharp, w_k^\sharp) \end{bmatrix} \right) \tag{B.7.5}$$

And we extend it bilinearly to all $k-$ forms.

Lemma B.7.2. ***(A formula for*** $g(v_1^\sharp, v_2^\sharp)$***)*** *Let* $v_1 = \sum_{i=1}^n v_1^i dx^i$ *and* $v_2 = \sum_{j=1}^n v_2^j dx^j$ *be* $1-$*forms. We then have that*

$$g(v_1^\sharp, v_2^\sharp) = \sum_{i,j=1}^n v_1^i g^{ij} v_2^j \tag{B.7.6}$$

Proof. $v_1^\sharp = \sum_{q=1}^n (\sum_{r_1=1}^n g^{qr_1} v_1^{r_1}) \frac{\partial}{\partial x^q}$ and $v_2^\sharp = \sum_{s=1}^n (\sum_{r_2=1}^n g^{sr_2} v_2^{r_2}) \frac{\partial}{\partial x^s}$. Thus we have

$$g(v_1^\sharp, v_2^\sharp) = \sum_{r_1=1}^n \sum_{r_2=1}^n v_1^{r_1} v_2^{r_2} (\sum_{q=1}^n \sum_{s=1}^n g^{qr_1} g^{sr_2} g_{qs}) = \sum_{r_1=1}^n \sum_{r_2=1}^n v_1^{r_1} g^{r_1 r_2} v_2^{r_2}$$

Which is the desired result.

■

Orientability

Definition B.7.3. (Orientable Manifold) A differentiable manifold M is said to be orientable if there exists an atlas $\mathscr{A}$ such that for any two charts (U, φ) and (V, ψ) in $\mathscr{A}$, the Jacobian determinant of the transition maps $\psi \circ \varphi^{-1}$ is positive. If we denote our charts as $x = (x^1, x^2, \cdots, x^n)$ and $y = (y^1, y^2, \cdots, y^n)$ respectively, we can suggestively then write $\det(\frac{\partial x^i}{\partial y^j}) > 0$

Note that, if x and y are coordinate charts around a point p, T_pM has bases $\{\frac{\partial}{\partial x^i}\}_{i \leq n}$ and $\{\frac{\partial}{\partial y^j}\}_{j \leq n}$. Chain rule lets us write $\frac{\partial}{\partial x^i} = \sum_{j=1}^n \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}$, and hence the transformation A that takes the basis $\{\frac{\partial}{\partial x^i}\}_{i \leq n}$ to $\{\frac{\partial}{\partial y^j}\}_{j \leq n}$ is of positive determinant.

Orthonormal frames and co-frames

Choose a frame $E_1, E_2, \cdots, E_n$ (from Gram-Schmidt process) that forms an orthonormal basis with respect to g, i.e. $g(E_i, E_j) = \delta_{ij}$. Consider the co-frame $\xi^1, \xi^2, \cdots, \xi^n$ dual to $E_1, E_2, \cdots, E_n$ in T_p^*M. $(\xi^i)^\sharp = \sum_j \xi_i^j E_j = \sum_j g(\xi^{i\sharp}, E_j) E_j$. Note that $g(\xi^{i\sharp}, E_j) = \xi^{i\sharp^\flat}(E_j) = \xi^i(E_j) = \delta_{ij}$ which gives us that $\xi^{i\sharp} = E_i$, and hence we have

$$\langle (\xi^i)^\sharp, (\xi^j)^\sharp \rangle_g = \delta_{ij}$$

Which tells us that the co-frame is also orthonormal with respect to the natural inner product $\langle \cdot, \cdot \rangle_g$. We collect this result in the following lemma:

Lemma B.7.4. *If* $E_1, E_2, \cdots, E_n$ *is an orthonormal frame with respect to* g*, then the dual co-frame* $\xi^1, \xi^2, \cdots, \xi^n$ *is also orthonormal with respect to the natural inner product* $\langle \cdot, \cdot \rangle_g$*.*

These ξ^i in B.7.4 form an orthonormal basis for T_p^*M, and hence the collection $\{\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k} : 1 \leq i_1 < i_2 < \cdots < i_k \leq n\}$ forms a basis for $\Lambda^k(T_pM)$. Let $\alpha = \sum_{I \in \mathscr{I}_k} \alpha^I \xi^I$ and $\beta = \sum_{I \in \mathscr{I}_k} \beta^I \xi^I$ be two $k-$forms written in this basis. Then we have

$$\langle \alpha, \beta \rangle_g = \sum_{I \in \mathscr{I}_k} \sum_{J \in \mathscr{I}_k} \langle \xi^I, \xi^J \rangle_g \alpha^I \beta^J = \sum_{I \in \mathscr{I}_k} \alpha^I \beta^I \tag{B.7.7}$$

The Volume Form

Let M be an orientable manifold and let $E_1, E_2, \cdots, E_k$ be a positively oriented orthonormal frame with respect to g. We define the volume form as the unique n form such that:

$$\text{vol}_g(E_1, E_2, \cdots, E_n) = 1 \tag{B.7.8}$$

Let $F_1, F_2, \cdots, F_n$ be another positively oriented (not necessarily orthonormal) frame. Suppose $\{E_i\}$ are related to $\{F_j\}$ via the orthogonal transformation A, i.e. $F_i = \sum_j A_i^j E_j$. Then we have

$$\text{vol}_g(F_1, F_2, \cdots, F_n) = \det(A) \tag{B.7.9}$$

$\det(A)$ can be found as follows:

$$g(F_i, F_k) = g(\sum_i A_j^i E_i, \sum_m A_k^m E_m) = \sum_i \sum_m A_j^i A_k^m \delta_{im} = \sum_i A_j^i A_k^i = (A^T A)_{jk}$$

Therefore,

$$\sqrt{\det(\{g(F_i, F_k)\}_{i,k \leq n})} = \det(A)$$

Let $x = (x^1, x^2, \cdots, x^n)$ and $y = (y^1, y^2, \cdots, y^n)$ be two coordinate charts around $p \in M$, such that $\{\frac{\partial}{\partial x^i}\}_{i \leq n}$ and $\{\frac{\partial}{\partial y^j}\}_{j \leq n}$ are both positively oriented. We then have that

$$\text{vol}_g(\frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \cdots, \frac{\partial}{\partial x^n}) = \sqrt{\det\left((g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}))\right)} \tag{B.7.10}$$

And

$$\text{vol}_g = \sqrt{\det\left((g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}))\right)} dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n \tag{B.7.11}$$

Hodge Star on Riemannian Manifolds

Definition B.7.5. (Hodge Star for Riemannian Manifolds) Let $\beta \in \Lambda^k$ be a $k-$form on an orientable Riemannian manifold (M, g). We define the Hodge Star $\star\beta$ as the *unique* $n-k$ form such that for every test k form α the following holds:

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \text{vol}_g \tag{B.7.12}$$

Lemma B.7.6. *Let $E_1, E_2, \cdots, E_n$ be a given positively oriented orthonormal frame. Let $\xi^1, \xi^2, \cdots, \xi^n$ be the dual co-frame. Then*

$$\star(\xi^{i_1} \wedge \xi^{i_2} \wedge \cdots \wedge \xi^{i_k}) = \text{sign}(R(I, I^c))\xi^{j_1} \wedge \xi^{j_2} \wedge \cdots \wedge \xi^{j_{n-k}} \tag{B.7.13}$$

Where $R(I, I^c)$ is the perumtation that rearranges $\{I, I^c\}$ to its natural order, and $j_1, j_2, \cdots, j_{n-k}$ are the canonically ordered incides of I^c.

Proof. The proof follows from the following observations:

1. The volume form is $\xi^1 \wedge \xi^2 \wedge \cdots \wedge \xi^n$
2. $\text{sign}(R(I, I^c))\xi^{i_1} \wedge \xi^{i_2} \cdots \wedge \xi^{i_k} \wedge \xi^{j_1} \wedge \cdots \wedge \xi^{j_{n-k}} = \text{vol}_g$.
3. $\langle \xi^{i_1} \wedge \cdots \wedge \xi^{i_k}, \xi^{i_1} \wedge \cdots \wedge \xi^{i_k} \rangle_g = 1$

■

Theorem B.7.7. ***(Double Star Identity)*** *Let f be a $k-$form. Then*

$$\star\star f = (-1)^{k(n-k)f} \tag{B.7.14}$$

Proof. Because the star is linear, it is enough to prove the identity on basis elements. Let $E_1, \cdots, E_n$ be a positively oriented orthonormal frame and $\xi^1, \cdots, \xi^n$ be its dual co-frame. $\star\star(\xi^{i_1} \wedge \xi^{i_2} \cdots \wedge \xi^{i_k}) = \text{sign}(R(I, I^c)) \star (\xi^{I^c}) = \text{sign}(R(I, I^c))\text{sign}(R(I^c, I))\xi^I$. The proof is done once we see that the product $\text{sign}(R(I, I^c))\text{sign}(R(I^c, I))$ is the sign of the permutation that takes swaps the first k terms in $[I, I^c]$ behind the last $n-k$ terms, which amounts to a sign of $(-1)^{k(n-k)}$.

■

Theorem B.7.8. ***(Hodge Star is an isometry)*** *The Hodge Star $\star : \Lambda^k \to \Lambda^{n-k}$ is an isometric isomorphism between Normed Linear Spaces.*

Proof. Note that the Hodge Star is linear. Since the dimensions of Λ^k and Λ^{n-k} are the same, it is enough to show that $\star$ is an isometry, i.e. $\langle \star\alpha, \star\beta\rangle_{g,n-k} = \langle \alpha, \beta\rangle_{g,k}$. Let $\xi^1, \xi^2, \cdots, \xi^n$ be the co-frame of the orthonormal frame $E_1, E_2, \cdots, E_n$. We saw in B.7.4 that these too, are orthonormal with respect to the natural inner product. Let $\alpha = \sum_{I\in\mathscr{I}_k} \alpha^I \xi^I$ and $\beta = \sum_{I\in\mathscr{I}_k} \beta^I \xi^I$. From B.7.7 we have $\langle \alpha, \beta\rangle_g = \sum_{I\in\mathscr{I}_k} \alpha^I \beta^I$. $\langle \star\alpha, \star\beta\rangle = \star\alpha \wedge \star\star\beta = (-1)^{k(n-k)} \star\star\beta \wedge \star\alpha = (-1)^{k(n-k)} \langle \alpha, \star\star\beta\rangle_g = \langle \alpha, \beta\rangle_g$ from the double star identity B.7.7.

■

Interior Product and Hodge Star

Definition B.7.9. (Interior Product) Let M be an oriented manifold. We define *the interior product* as the adjoint of the wedge product with respect to the natural inner product, i.e. for every $\alpha \in \Lambda^k$, $\beta \in \Lambda^l$ and $\eta \in \Lambda^{l-k}$

$$\langle \alpha \wedge \eta, \beta\rangle_g = \langle \eta, \alpha \lrcorner \beta\rangle_g \tag{B.7.15}$$

Lemma B.7.10. *Let $E_1, E_2, \cdots, E_n$ be a positively oriented orthonormal basis and $\xi^1, \xi^2, \cdots, \xi^n$ be its dual co-frame. We then have:*

$$\xi^I \lrcorner \xi^J = \begin{cases} 0 & \text{if } I \not\subseteq J \\ \operatorname{sign}(R(I, J \setminus I))\xi^{J\setminus I} & \text{if } I \subseteq J \end{cases}$$

Proof. The proof is the same as the euclidean case B.5.2, since the co-frame is orthonormal with respect to the inner product $\langle \cdot, \cdot\rangle_g$.

■

Theorem B.7.11. *Let $\alpha \in \Lambda^k$ $\beta \in \Lambda^l$. We then have $\alpha \lrcorner \beta = (-1)^{(n-l)(l-k)} \star (\alpha \wedge \star\beta)$*

Proof. The proof is the same as the euclidean case B.6.3.

■

Theorem B.7.12. *Let X be a vector field and $\omega \in \Lambda^k$ and $\eta \in \Lambda^l$. We then have*

$$\imath_X(\omega \wedge \eta) = \imath_X(\omega) \wedge \eta + (-1)^k \omega \wedge \imath_X(\eta) \tag{B.7.16}$$

Corollary B.7.13. Let $\xi \in \Lambda^1$ and $\eta \in \Lambda^k$. Then we have the following identity:

$$||\xi||^2 \eta = \xi \wedge (\xi \lrcorner \eta) + \xi \lrcorner (\xi \wedge \eta) \tag{B.7.17}$$

B.8 Pullback of Forms

Let $F : U \to V$, where $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$. If F is smooth, the Frechet derivative at p is defined as the unique linear map $DF_p : \mathbb{R}^n \to \mathbb{R}^m$ satisfying

$$\lim_{|h|\to 0} \frac{||F(p+h) - F(p) - DF_p(h)||_m}{||h||_n} = 0$$

We have $DF_p : T_pU \to T_{f(p)}V$ where $DF_p(v) \in T_{f(p)}V$ such that

$$DF_p(v)(g) = v(g \circ F)$$

Where $g \in C^{\infty}_{f(p)}(V)$. In coordinates we have

$$DF_p(\frac{\partial}{\partial x^i}|_p) = \sum_{j=1}^{m} DF_p(\frac{\partial}{\partial x^i}|_p)(y^j)\frac{\partial}{\partial y^j}|_{F(p)} \tag{B.8.1}$$

which gives

$$DF_p(v) = DF_p(\sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}|_p) = \sum_{i=1}^{n}\sum_{j=1}^{m} DF_p(\frac{\partial}{\partial x^i}|_p)(y^j)v^i\frac{\partial}{\partial y^j}|_{F(p)} = \sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\partial}{\partial x^i}|_p(y^j \circ F)v^i\frac{\partial}{\partial y^j}|_{F(p)}$$

If we define $F^j = y^j \circ F$ (The $j-$th coordinate of F) we get

$$DF_p(v) \equiv \sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\partial F^j}{\partial x^i}|_p v^i \frac{\partial}{\partial y^j}|_{F(p)}$$

Where $\{\frac{\partial F^j}{\partial x^i}|_p : 1 \le i \le n, 1 \le j \le m\}$ is the Jacobian Matrix.

Definition B.8.1. (Pullback of a $k-$form) Let $\omega : V \to \amalg_{q\in V}\Lambda^k(T_q(V))$ be a $k-$form on $V \subset \mathbb{R}^m$. We can get a $k-$form on $U \subseteq \mathbb{R}^n$ via F as $F^*\omega_p \in \Lambda^k(T_p(U))$ given by

$$F^*\omega_p(v_1, v_2, \cdots, v_k) = \omega_{f(p)}\big(DF_p(v_1), DF_p(v_2), \cdots, DF_p(v_k)\big)$$

B.8.1 Pullback of a $1-$form

Let $\omega = dy^q$ be a basis $1-form$ in $V \subset \mathbb{R}^m$.

$$F^*(dy^q)_p(v) = dy^q_{F(p)}(DF_p(v))$$

Which gives

$$dy^q_{F(p)}(\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\partial F^j}{\partial x^i}|_p v^i\frac{\partial}{\partial y^j}|_{F(p)})$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\partial F^j}{\partial x^i}|_p v^i\delta_{jq} = \sum_{i=1}^{n}\frac{\partial F^q}{\partial x^i}|_p v^i = \sum_{i=1}^{n}\frac{\partial F^q}{\partial x^i}|_p dx^i_p(v)$$

Therefore we can set up the equivalence

$$F^*(dy^q)_p \equiv \sum_{i=1}^{n}\frac{\partial F^q}{\partial x^i}|_p dx^i_p \tag{B.8.2}$$

Note that $F^*y^q = y^q \circ F \equiv F^q$, and its exterior derivative

$$dF^q = d(F^*(y^q)) = (\sum_{i=1}^{n}\frac{\partial}{\partial x^i}_p dx^i) \wedge (F^*(y^q)) = \sum_{i=1}^{n}\frac{\partial y^q \circ F}{\partial x^i}|_p dx^i_p$$

And hence we see that for basis $1-$forms

$$d(F^*(y^q)) = F^*(dy^q)$$

If $\omega : V \to \amalg_{q\in V}T^*_q(V)$ is any general one form given by

$$\omega \equiv \sum_{j=1}^{m}\omega^j dy^j$$

then we have

$$F^*(\omega)_p(v) := \omega_{F(p)}(DF_p(v)) = \sum_{j=1}^{m}\omega^j dy^j_{F(p)}(DF_p(v))$$

which yields

$$\sum_{j=1}^{m}\omega^j(F(p))\sum_{i=1}^{n}\frac{\partial F^j}{\partial x^i}|_p dx^i_p(v) = \sum_{j=1}^{m}\omega^j_{(F(p))}dF^j_p(v)$$

which gives us an expansion of the form

$$F^*(\omega)_p \equiv \sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\partial F^j}{\partial x^i}|_p\omega^j_{F(p)}dx^i_p \equiv \sum_{j=1}^{m}\omega^j_{(F(p))}dF^j_p$$

Yet another suggestive way to write this would be

$$F^*(\omega)_p = F^*(\sum_{j=1}^{m}\omega^j dy^j)_p = \sum_{j=1}^{m}(\omega^j\circ F)_{(p)}dF^j_p$$

B.8.2 Pullback of a $k-$form

Let $\omega = dy^I = dy^{i_1}\wedge dy^{i_2}\cdots\wedge dy^{i_k}\in\Omega(V,\Lambda^k)$ be a k form on $V\subset\mathbb{R}^m$.

$$F^*(dy^I)_p(\sum_{j_1=1}^{n}v_1^{j_1}\frac{\partial}{\partial x^{j_1}}|_p,\sum_{j_2=1}^{n}v_2^{j_2}\frac{\partial}{\partial x^{j_2}}|_p,\cdots,\sum_{j_k=1}^{n}v_k^{j_k}\frac{\partial}{\partial x^{j_k}}|_p)$$

$$=\sum_{1\le j_1,j_2,\cdots,j_k\le n}(v_1^{j_1}v_2^{j_2}\cdots v_k^{j_k})F^*(dy^I)_p(\frac{\partial}{\partial x^{j_1}}|_p,\frac{\partial}{\partial x^{j_2}}|_p,\cdots,\frac{\partial}{\partial x^{x^{j_k}}}|_p) \tag{B.8.3}$$

$$F^*(dy^I)_p(\frac{\partial}{\partial x^{j_1}}|_p,\frac{\partial}{\partial x^{j_2}}|_p,\cdots,\frac{\partial}{\partial x^{j_k}}|_p,) = dy^I_{F(p)}(DF_p(\frac{\partial}{\partial x^{j_1}}|_p),DF_p(\frac{\partial}{\partial x^{j_2}}|_p),\cdots,DF_p(\frac{\partial}{\partial x^{j_k}}|_p))$$

From B.8.1 we see that

$$F^*(dy^I)(\frac{\partial}{\partial x^{j_1}}|_p,\frac{\partial}{\partial x^{j_2}}|_p,\cdots,\frac{\partial}{\partial x^{j_k}}|_p) = dy^I_{F(p)}\Big(\sum_{q_1=1}^{m}\frac{\partial F^{q_1}}{\partial x^{j_1}}|_p\frac{\partial}{\partial y^{q_1}}|_{F(p)},\sum_{q_2=1}^{m}\frac{\partial F^{q_2}}{\partial x^{j_2}}|_p\frac{\partial}{\partial y^{q_2}}|_{F(p)},\cdots,\sum_{q_k=1}^{m}\frac{\partial F^{q_k}}{\partial x^{j_k}}|_p\frac{\partial}{\partial y^{q_k}}|_{F(p)}\Big)$$

$$=\sum_{1\le q_1,q_2,\cdots,q_k\le m}\Big(\frac{\partial F^{q_1}}{\partial x^{j_1}}|_p\cdot\frac{\partial F^{q_2}}{\partial x^{j_2}}|_p\cdots\frac{\partial F^{q_k}}{\partial x^{j_k}}|_p\Big)dy^I_{F(p)}(\frac{\partial}{\partial y^{q_1}}|_{F(p)},\frac{\partial}{\partial y^{q_2}}|_{F(p)},\cdots,\frac{\partial}{\partial y^{q_k}}|_{F(p)})$$

$$=\sum_{1\le q_1,q_2,\cdots,q_k\le m}\Big(\frac{\partial F^{q_1}}{\partial x^{j_1}}|_p\cdot\frac{\partial F^{q_2}}{\partial x^{j_2}}|_p\cdots\frac{\partial F^{q_k}}{\partial x^{j_k}}|_p\Big)\det\Big((\delta^{q_t}_{i_s}):1\le s,t\le k\Big)$$

Combining this with B.8.3 gives us

$$F^*(dy^I)(v_1,\cdots,v_k)=\sum_{\substack{1\le j_1,j_2,\cdots,j_k\le n\\1\le q_1,q_2,\cdots,q_k\le m}}(v_1^{j_1}v_2^{j_2}\cdots v_k^{j_k})\Big(\frac{\partial F^{q_1}}{\partial x^{j_1}}|_p\cdot\frac{\partial F^{q_2}}{\partial x^{j_2}}|_p\cdots\frac{\partial F^{q_k}}{\partial x^{j_k}}|_p\Big)\det\Big((\delta^{q_t}_{i_s})_{1\le s,t\le k}\Big) \tag{B.8.4}$$

$$=\sum_{1\le j_1,j_2,\cdots,j_k\le n}(v_1^{j_1}v_2^{j_2}\cdots v_k^{j_k})\sum_{\sigma\in sym(k)}\operatorname{sign}(\sigma)\Big(\frac{\partial F^{i_{\sigma(1)}}}{\partial x^{j_1}}|_p\cdot\frac{\partial F^{i_{\sigma(2)}}}{\partial x^{j_2}}|_p\cdots\frac{\partial F^{i_{\sigma(k)}}}{\partial x^{j_k}}|_p\Big)$$

$$=\sum_{1\le j_1,\cdots,j_k\le n}\Big(dx^{j_1}(v_1)dx^{j_2}(v_2)\cdots dx^{j_k}(v_k)\Big)\det\big((\frac{\partial F^{i_r}}{\partial x^{j_s}}|_p)_{1\le r,s\le k}\big)$$

$$=\sum_{1\le j_1<j_2<\cdots<j_k\le n}\sum_{\sigma\in sym(k)}\Big(dx^{j_{\sigma(1)}}(v_1)dx^{j_{\sigma(2)}}(v_2)\cdots dx^{j_{\sigma(k)}}(v_k)\Big)\det\big((\frac{\partial F^{i_r}}{\partial x^{j_{\sigma(s)}}}|_p)_{1\le s,r\le k}\big)$$

$$=\sum_{1\le j_1<j_2<\cdots<j_k\le n}\det\big((\frac{\partial F^{i_r}}{\partial x^{j_s}}|_p)_{1\le r,s\le k}\big)\sum_{\sigma\in sym(k)}\operatorname{sign}(\sigma)dx^{j_{\sigma(1)}}(v_1)dx^{j_{\sigma(2)}}(v_2)\cdots dx^{j_{\sigma(k)}}(v_k)$$

Which finally yields

$$F^*(dy^I)_p=\sum_{1\le j_1<j_2<\cdots<j_k\le n}\det\big((\frac{\partial F^{i_r}}{\partial x^{j_s}}|_p)_{1\le r,s\le k}\big)dx^{j_1}_p\wedge dx^{j_2}_p\cdots dx^{j_k}_p \tag{B.8.5}$$

We now look at $d(F^*y^{i_1})_p\wedge d(F^*y^{i_2})_p\cdots\wedge d(F^*y^{i_k})_p = dF^{i_1}\wedge dF^{i_2}\cdots dF^{i_k}$.

$$=\Big(\sum_{j_1=1}^{n}\frac{\partial F^{i_1}}{\partial x^{j_1}}|_p dx^{j_1}\Big)\wedge\Big(\sum_{j_2=1}^{n}\frac{\partial F^{i_2}}{\partial x^{j_2}}|_p dx^{j_2}\Big)\wedge\cdots\wedge\Big(\sum_{j_k=1}^{n}\frac{\partial F^{i_k}}{\partial x^{j_k}}|_p dx^{j_k}\Big)$$

$$= \sum_{1\leq j_1,j_2,\cdots,j_k\leq n} \Big(\frac{\partial F^{i_1}}{\partial x^{j_1}}|_p \cdot \frac{\partial F^{i_2}}{\partial x^{j_2}}|_p \cdots \frac{\partial F^{i_k}}{\partial x^{j_k}}|_p\Big) dx^{j_1} \wedge \cdots \wedge dx^{j_k}$$

$$= \sum_{1\leq j_1<j_2<\cdots<j_k\leq n} dx^{j_1} \wedge dx^{j_2} \cdots \wedge dx^{j_k} \sum_{\sigma\in sym(k)} \text{sign}(\sigma)\Big(\frac{\partial F^{i_1}}{\partial x^{j_{\sigma(1)}}}|_p \cdot \frac{\partial F^{i_2}}{\partial x^{j_{\sigma(2)}}}|_p \cdots \frac{\partial F^{i_k}}{\partial x^{j_{\sigma(k)}}}|_p\Big)$$

$$= \sum_{1\leq j_1<j_2<\cdots<j_k\leq n} dx^{j_1} \wedge \cdots \wedge dx^{j_k} \det\Big((\frac{\partial F^{i_r}}{\partial x^{j_s}}|_p)_{1\leq r,s\leq k}\Big) \tag{B.8.6}$$

Which agrees with B.8.5. Hence we have the following identity:

$$F^*(dy^{i_1} \wedge dy^{i_2} \wedge \cdots \wedge dy^{i_k}) = d(F^*y^{i_1}) \wedge d(F^*y^{i_2}) \wedge \cdots \wedge d(F^*y^{i_k}) \tag{B.8.7}$$

Corollary B.8.2. If $F : U_{\subset\mathbb{R}^n} \to V_{\subset\mathbb{R}^n}$ is a smooth Diffeomorphism, and coordinates for U are given by $x^1, x^2, \cdots, x^n$, and those for V given by $y^1, y^2, \cdots, y^n$, we have a change of variables formula for $k-$forms (In view of formula B.8.5) given by:

$$F^*(dy^1 \wedge dy^2 \wedge \cdots \wedge dy^n)_p = \det\Big((\frac{\partial F^j}{\partial x^i})_{1\leq i,j\leq n}\Big) dx^1 \wedge dx^2 \wedge \cdots dx^n$$

Remark B.8.3. The proof of B.8.2 is simply an application of B.8.5. This is the **change of variables formula** for *top forms*.

Theorem B.8.4. ***(Pullback of Wedge)*** *Let $\alpha \in \Omega(V, \Lambda^k)$ be a $k-$form on $V \subset \mathbb{R}^m$ and let $\beta \in \Omega(V, \Lambda^l)$ be an $l-$form on V. For smooth function $F : U \subset \mathbb{R}^n \to V$ we have*

$$F^*(\alpha \wedge \beta) = F^*(\alpha) \wedge F^*(\beta)$$

Proof. Recall $F^*(\alpha \wedge \beta)_p(v_1, v_2, \cdots, v_k, v_{k+1}, \cdots, v_{k+l}) = (\alpha \wedge \beta)_{F(p)}\big(DF_p(v_1), \cdots, DF_p(v_{k+l})\big) =$

$$\sum_{\sigma\in sym(k,l)} \text{sign}(\sigma)\alpha_{F(p)}(DF_p(v_{\sigma(1)}), DF_p(v_{\sigma(2)}), \cdots, DF_p(v_{\sigma(k)}))\beta_{F(p)}(DF_p(v_{\sigma(k+1)}), \cdots, DF_p(v_{\sigma(k+l)}))$$

$$= \sum_{\sigma\in sym(k,l)} \text{sign}(\sigma)F^*(\alpha)_p(v_{\sigma(1)}, \cdots, v_{\sigma(k)})F^*(\beta)_p(v_{\sigma(k+1)}, \cdots, v_{\sigma(k+l)}) = F^*(\alpha) \wedge F^*(\beta)$$

■

If $\omega = \sum_{I\in\mathscr{T}_k} \omega^I dy^I$ is a general $k-$form on $V \subseteq \mathbb{R}^m$, then we have

$$F^*(\sum_I \omega^I dy^I)_p = \sum_I \omega^I_{F(p)} dF^I_p \tag{B.8.8}$$

Where $dF^I_p = dF^{i_1}_p \wedge dF^{i_2}_p \wedge \cdots \wedge dF^{i_k}_p$.

Lemma B.8.5. ***(F^* commutes with d)*** *Let $F : U_{\subset\mathbb{R}^n} \to V_{\subset\mathbb{R}^m}$ be smooth, and ω a $k-$form on V. We then have that*

$$F^*(d\omega) = d(F^*\omega)$$

Proof. If $\omega = \sum_{I\in\mathscr{T}_k} \omega^I dy^I$ From B.8.8 we have that

$$F^*(\omega)_p = \sum_I \omega^I_{F(p)} dF^{i_1}_p \wedge dF^{i_2}_p \cdots \wedge dF^{i_k}_p$$

When we apply the exterior derivative to this expression we have (and recalling the antiderivation property)

$$d(F^*(\omega))_p = \sum_I d(\omega^I \circ F)_p \wedge dF^{i_1}_p \wedge dF^{i_2}_p \cdots dF^{i_k}_p + (\omega^I \circ F)d(dF^{i_1} \wedge dF^{i_2} \cdots \wedge dF^{i_k})_p$$

The second term vanishes because $d^2 = 0$. This gives

$$d(F^*(\omega))_p = \sum_I \sum_{i=1}^n \frac{\partial(\omega^I \circ F)}{\partial x^i}|_p dx_p^i \wedge dF_p^{i_1} \wedge dF_p^{i_2} \wedge \cdots \wedge dF_p^{i_k} = \sum_I d(\omega^I \circ F) \wedge dF_p^I$$

Now, if $\omega = \sum_I \omega^I dy^I$, we have $d\omega = \sum_I d\omega^I \wedge dy^I$. Pulling this back gives us

$$F^*(d\omega) = \sum_I F^*(d\omega^I) \wedge F^*(dy^I) = \sum_I d(\omega^I \circ F) \wedge dF^I$$

Which establishes equality.

■

Pullback and Interior Product

Remark B.8.6. For a smooth map $F\colon M \to N$, the identity $i_X(F^*\omega) = F^*(i_Y\omega)$ is ill-defined for generic vector fields since X is a vector field from M and Y is a vector field on N. Naturality under pullbacks holds precisely when X and Y are F-related, which motivates the next definition.

Definition B.8.7. ($F-$related Vector Fields) Let $F : M \to N$ be smooth. Vector fields X on M is said to be **F-related** to a vector field Y on N if

$$dF_p(X_p) = Y_{F(p)}$$

for every $p \in M$.

Theorem B.8.8. *Let $F\colon M \to N$ be a smooth map. If X is a vector field on M and Y on N are F-related, then for any differential form $\omega \in \Omega^k(N)$,*

$$i_X(F^*\omega) = F^*(i_Y\omega).$$

Proof. Evaluating the left-hand side at $p \in M$ on vectors $v_1, \ldots, v_{k-1} \in T_pM$ yields:

$$\begin{aligned}(i_X F^*\omega)_p(v_1, \ldots, v_{k-1}) &= (F^*\omega)_p(X_p, v_1, \ldots, v_{k-1}) \\ &= \omega_{F(p)}(dF_pX_p, dF_pv_1, \ldots, dF_pv_{k-1}) \\ &= \omega_{F(p)}(Y_{F(p)}, dF_pv_1, \ldots, dF_pv_{k-1}) \\ &= (F^*i_Y\omega)_p(v_1, \ldots, v_{k-1}).\end{aligned}$$

■

Lemma B.8.9. *Let $U \subseteq \mathbb{R}^n$ be an open connected set and $f : U \to \mathbb{R}$ be a smooth function. Then $df = 0$ if and only if f is constant on U.*

Proof. $\implies$) Let f be constant on U. Then $\frac{\partial f}{\partial x^i} = 0$ for all $i \leq n$, and hence $df_p = \sum_{i=1}^n \frac{\partial f}{\partial x^i}|_p dx_p^i = 0$.

$\impliedby$) Say $df_p = 0$ for all $p \in U$. Let x and y be arbitrary fixed points in U and let $c : [a, b] \to U$ be any smooth curve connecting x and y (Such a curve exists because U is path-connected, and hence continuity is guarenteed. Smoothening of the polygon edges with smooth curves guarentees smoothness). Note that $c^*f \equiv f \circ c : [a, b] \to \mathbb{R}$ is a smooth 0 form on $[a, b]$. $d(c^*f) = d(f \circ c) = \frac{d(f \circ c(t))}{dt} dt = c^*df = 0$ (from B.8.5). Therefore $d(c^*f) = 0$, so $f \circ c(a) = f \circ c(b)$ which gives $f(x) = f(y)$.

■

B.9 Manifolds with Boundary

PENDING We ask the question: What does it mean for a function to be smooth on an *arbitrary* set $S \subset \mathbb{R}^n$

B.10 Integration of 1−forms

Let $U \subseteq \mathbb{R}^n$ be an open set and α be a 1−form.

Definition B.10.1. (Parametrized Paths) Any map $c : [a, b] \to U \subset \mathbb{R}^n$ that is smooth is said to be a *path* from $c(a)$ to $c(b)$.

The pullback of α given by $c^*\alpha$ is a 1−form on $[a, b]$. Suppose $c^*\alpha = hdt$ where t is the coordinate function of $[a, b]$ and h a real valued smooth function.

Definition B.10.2. (Integral of a 1−form over a curve)

$$\int_c \alpha = \int_{[a,b]} c^*\alpha = \int_a^b h(t)dt \tag{B.10.1}$$

where the right most equality makes sense in the standard way. Moreover, if $\alpha = \sum_{i=1}^n f^i dx^i$, we have $c^*(\sum_{i=1}^n f^i dx^i) = \sum_{i=1}^n c^*(f^i)d(c^*x^i) = \sum_{i=1}^n (f^i \circ c)\frac{dc^i}{dt}dt$ where $c^i = x^i \circ c$. Its integral would then be

$$\int_c \alpha = \int_a^b c^*\alpha = \sum_{i=1}^n \int_a^b f^i \circ c(t)\frac{dc^i}{dt}(t)dt = \int_a^b \sum_{i=1}^n f^i(c(t))(c^i)'(t)dt \tag{B.10.2}$$

Which can be suggestively written as

$$\int_a^b \alpha^\sharp(c(t)) \cdot c'(t)dt \tag{B.10.3}$$

where $\alpha^\sharp$ is the vector field associated with the 1−form α via the inner product.

B.11 The Stokes', Green's and Gauss' Theorems

Definition B.11.1. (Lipschitz and C^k Set) Let $\Omega \subseteq \mathbb{R}^n$ be an open domain and let $\partial\Omega$ denote its boundary. Ω is said to be a *Lipschitz set* (C^k set) if for every point $p \in \partial\Omega$ there exists a hyperplane H of dimension $n-1$ passing through p, a Lipschitz continuous (C^k) function $g : H \to \mathbb{R}$ and reals $r > 0, h > 0$ such that:

1. $\Omega \cap C = \{x + y\hat{n} \ : \ x \in B_r(p) \cap H \text{ and } -h < y < g(x)\}$
2. $\partial\Omega \cap H = \{x + y\hat{n} \ : \ x \in B_r(p) \cap H \text{ and } y = g(x)\}$

Where

1. $\hat{n}$ is the unit normal to H.
2. $B_r(p)$ is the ball of radius r at p.
3. $C := \{x + y\hat{n} : x \in B_r(p) \cap H \text{ and } -h < y < h\}$ (Cylinder of height $2h$ and radius r at p)

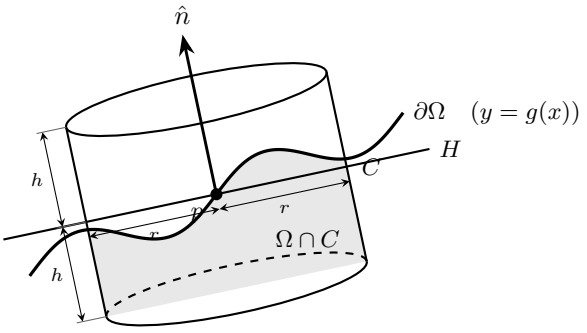


Figure B.1: Cylinder around a boundary point $p \in \partial\Omega$ from definition B.11.1

B.12 The Codifferential δ

B.12.1 Motivation

Let $\alpha \in \Omega\Lambda^{k-1}$ and $\beta \in \Omega\Lambda^k$. Here we can invoke the definition of the Hodge star (B.6.1) to get

$$\langle d\alpha, \beta\rangle vol = d\alpha \wedge (\star\beta)$$

$$\int_\Omega \langle d\alpha, \beta\rangle vol = \int_\Omega \Big(d\alpha \wedge (\star\beta) + (-1)^{k-1}\alpha \wedge d(\star\beta) + (-1)^k \alpha \wedge d(\star\beta)\Big)$$

Using the Antiderivation identity B.3.12 we get

$$\int_\Omega \langle d\alpha, \beta\rangle vol = \int_\Omega \Big(d(\alpha \wedge \star\beta) + (-1)^k \alpha \wedge d(\star\beta)\Big) \tag{B.12.1}$$

$$\int_\Omega \langle d\alpha, \beta\rangle vol = \Big(\int_{\partial\Omega} \alpha \wedge (\star\beta) + (-1)^k \int_\Omega \alpha \wedge d(\star\beta)\Big)$$

Since $\star\beta$ is a $n-k$ form and $d(\star\beta)$ is an $n-(k-1)$ form, we apply the definition of Hodge star again to replace $\alpha \wedge d(\star\beta)$ with $(-1)^{(n-(k-1))(k-1)}\langle \alpha, \star d \star \beta\rangle vol$ to get:

$$\int_\Omega \langle d\alpha, \beta\rangle vol = \Big(\int_{\partial\Omega} \alpha \wedge (\star\beta) + (-1)^k (-1)^{(n-(k-1))(k-1)} \int_\Omega \langle \alpha, \star d(\star\beta)\rangle vol\Big) \tag{B.12.2}$$

Note that (B.12.2) is a formula that is independent of the definition for *Codifferential* that will follow. This is important because different texts (cf. [Jos17] and [CDK12]) opt for different sign conventions for the Codifferential.

Definition B.12.1. (Codifferential) We define a map $\delta : \oplus_{k=0}^\infty \Omega\Lambda^k \to \oplus_{k=0}^\infty \Omega\Lambda^k$ given by

$$\delta(\omega) := (-1)^{n(k-1)} \star d \star (\omega) \tag{B.12.3}$$

for a $k-$form ω.

Using B.12.1 in (B.12.2) we immediately see that

$$\int_\Omega \langle d\alpha, \beta\rangle \mathrm{vol} = (-1) \int_\Omega \langle \alpha, \delta\beta\rangle \mathrm{vol} + \int_{\partial\Omega} \alpha \wedge \star\beta \tag{B.12.4}$$

This is called the *integration by parts formula* for forms.

Remark B.12.2. 1. Under the conventions of B.12.1, the formula B.12.4 takes the above form. But if we propose that $\delta = (-1)^{nk+n+1} \star d\star$, or $(-1)^{k(n-k+1)} \star d\star$, we see different possible factors in B.12.4.
2. If we assume that our set either has no boundary or that the forms vanish on the boundary, the formula B.12.4 yields

$$\int_\Omega \langle d\alpha, \beta\rangle + \int_\Omega \langle \alpha, \delta\beta\rangle = 0$$

B.12.2 Coordinate formulas for the Codifferential

Case 1: Let $\omega = dx^I$. Where $I \in \mathscr{T}_k(\mathbb{R}^n)$. Then $\delta(\omega) = (-1)^{n(k-1)} \star d \star (dx^I)$ gives us

$$(-1)^{n(k-1)} \operatorname{sgn}(R(I, I^c)) \star (d(dx^{I^c})) = 0$$

Case 2: Consider the case where $\omega = f dx^I$. For ease sake, let $(-1)^{n(k-1)} := \kappa$. We then have $\delta(\omega) = \kappa \star d \star (f dx^I) = \kappa \star d(f \operatorname{sgn}(R(I, I^c)) dx^{I^c})$ which gives

$$\kappa \operatorname{sgn}(R(I, I^c)) \star d(f dx^{I^c}) = \kappa \operatorname{sgn}(R(I, I^c)) \star \Big(\sum_{q=1}^n \frac{df}{dx^q} dx^q \wedge dx^{I^c}\Big)$$

since the summand is 0 for when $q \notin I$ we have

$$\delta(f dx^I) = \kappa \operatorname{sgn}(R(I, I^c)) \Big(\sum_{\gamma=1}^k \frac{df}{dx^{i_\gamma}} \star \big(dx^{i_\gamma} \wedge dx^{I^c}\big)\Big) \tag{B.12.5}$$

We will continue the derivation after the following proposition:

Proposition B.12.3. Let $I = (i_1, i_2, \cdots, i_k) \in \mathscr{T}_k(\mathbb{R}^n)$. Then

$$\star(dx^{i_\gamma} \wedge dx^{I^c}) = (-1)^{n-k+\gamma-1+k(n-k)}\text{sgn}(R(I, I^c))dx^{I_{\hat{i}_\gamma}} \tag{B.12.6}$$

Proof. $\star(dx^{i_\gamma} \wedge dx^{I^c}) = dx^{I_{\hat{i}_\gamma}}(R)$ where R is the coefficient such that $dx^{i_\gamma} \wedge dx^{I^c} \wedge dx^{I_{\hat{i}_\gamma}} = R\text{vol}$. Note that

$$dx^{i_\gamma} \wedge dx^{I^c} \wedge dx^{I_{\hat{i}_\gamma}} = (-1)^{1(n-k)}dx^{I^c} \wedge (-1)^{\gamma-1}dx^I$$

Which gives

$$(-1)^{n-k}(-1)^{\gamma-1}\text{sgn}(I, I^c)(-1)^{k(n-k)}\text{vol}$$

And hence $R = (-1)^{n-k+\gamma-1+k(n-k)}\text{sgn}(R(I, I^c))$

■

From the proposition, we have

$$\delta(fdx^I) = \kappa\text{sgn}(R(I, I^c))^2(-1)^{n-k+k(n-k)}\Big(\sum_{\gamma=1}^{k}(-1)^{\gamma-1}\frac{\partial f}{\partial x^{i_\gamma}}dx^{I_{\hat{i}_\gamma}}\Big)$$

$$\delta(fdx^I) = \Big(\sum_{\gamma=1}^{k}(-1)^{\gamma-1}\frac{\partial f}{\partial x^{i_\gamma}}dx^{I_{\hat{i}_\gamma}}\Big) \tag{B.12.7}$$

Remark B.12.4. Note that, if we adopted a different convention for δ (like $\delta = (-1)^{k(n-k+1)}$) we would then have the following formula instead of (B.12.7):

$$\delta(fdx^I) = (-1)^n\Big(\sum_{\gamma=1}^{k}(-1)^{\gamma-1}\frac{\partial f}{\partial x^{i_\gamma}}dx^{I_{\hat{i}_\gamma}}\Big) \tag{B.12.8}$$

General Case: Let $\omega = \sum_{I\in\mathscr{T}_k(\mathbb{R}^n)} f^I dx^I$ Then

$$\delta(\sum_{I\in\mathscr{T}_k} f^I dx^I) = \sum_{I\in\mathscr{T}_k}\delta(f^I dx^I) = \sum_{I\in\mathscr{T}_k}\sum_{\gamma=1}^{k}(-1)^{\gamma-1}\frac{\partial f^I}{\partial x^{i_\gamma}}dx^{I_{\hat{i}_\gamma}} \tag{B.12.9}$$

Alternative Formula for the Codifferential:

Let $\omega = \sum_{1\le i_1<\cdots<i_k\le n}\omega^{i_1 i_2\cdots i_k}dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k}$ and let $\delta(\omega) = \kappa \star d \star (\omega)$.

$$\delta(\omega) = \kappa \star d\Big(\sum_{1\le i_1<\cdots<i_k\le n}\omega^{i_1\cdots i_k}\text{sgn}(R(I, I^c))dx^{I^c}\Big)$$

$$= \kappa\star\Big(\sum_{1\le i_1<\cdots<i_k\le n}\text{sgn}(R(I, I^c))d\omega^{i_1\cdots i_k} \wedge dx^{I^c}\Big) = \kappa\star\Big(\sum_{1\le i_1<\cdots<i_k\le n}\text{sgn}(R(I, I^c))\Big(\sum_{q=1}^{n}\frac{\partial\omega^{i_1 i_2\cdots i_k}}{\partial x^q}dx^q\Big) \wedge dx^{I^c}\Big)$$

$$\delta(\omega) = \kappa\Big(\sum_{q=1}^{n}\sum_{I\in\mathscr{T}_{k-1}}\text{sgn}(R([qI], [qI]^c))\frac{\partial\omega^{[qI]}}{\partial x^q} \star dx^q \wedge dx^{[qI]^c}\Big) \tag{B.12.10}$$

A modification to the proof of B.12.6 yields us the following computation for $\star(dx^q \wedge dx^{[qI]^c})$:

$$\star(dx^q \wedge dx^{[qI]^c}) = (-1)^{n-k+\gamma-1+k(n-k)}\text{sgn}(R([qI], [qI]^c))dx^I \tag{B.12.11}$$

where $i_{\gamma-1} < q < i_\gamma$. Combining (B.12.11) and (B.12.10) we get:

$$\delta(\omega) = \kappa\Big(\sum_{q=1}^{n}\sum_{I\in\mathscr{T}_{k-1}}\text{sgn}(R([qI], [qI]^c))^2(-1)^{n-k+\gamma-1+k(n-k)}\frac{\partial\omega^{[qI]}}{\partial x^q}dx^I\Big)$$

which gives

$$\delta(\omega) = \kappa(-1)^{n-k+k(n-k)}\Big(\sum_{q=1}^{n}\sum_{I\in\mathscr{T}_{k-1}}\frac{\partial\omega^{q,I}}{\partial x^q}dx^I\Big)$$

Which simplifies to

$$\delta(\omega) = \Big(\sum_{q=1}^{n} \sum_{I \in \mathscr{T}_{k-1}} \frac{\partial \omega^{q,I}}{\partial x^q} dx^I \Big) \tag{B.12.12}$$

Where the notation $\omega^{q,I}$ is understood form the equation B.5.6.

Remark B.12.5. This is the formula persented in [CDS18] pg. 465, 2 (vii).

B.13 Boundary Calculus and Tangential and Normal Components

!!Not complete!!

1. tangential and normal part proofs
2. admissible boundary coordinates proof
3. subtleties with lipshitz domains and boundaries.

Appendix C

Measure Theory, Functional Analysis and Sobolev Spaces

C.1 Topological Vector Spaces

The following development is loosely based on [Rud91].

Definition C.1.1. (Topological Vector Space) A Vector Space X over $\mathbb{K}(\mathbb{R}$ or $\mathbb{C})$ is a *Topological Vector Space*(TVS, from now on) if it has a topology $\mathscr{T}_X$ such that:

1. Point sets are closed (T_1 axiom).
2. The vector space operations $(v, u) \mapsto v_u$ and $(\alpha, x) \mapsto \alpha x$ are continuous as maps from $X \times X \to X$ and $\mathbb{K} \times X \to X$ respectively.

Lemma C.1.2. *The translation map $T_a : X \to X$ given by $T_a(x) = a + x$, and the multiplication map $M_\lambda : X \to X$ given by $M_\lambda(x) := \lambda x$ are Homeomorphisms.*

C.1.2 tells us that U is open in X if and only if $U + a$ is open in X. This implies that the topology of a TVS is completely determined by a local base at $x = 0$.

Definition C.1.3. (Convex Set) A set $C \subset X$ is said to be *convex* if for any $x, y \in X$, $t \in [0, 1]$,

$$tx + (1 - t)y \in X$$

In other words, $tC + (1 - t)C \subset C$.

Definition C.1.4. (Locally Convex TVS (LCTVS)) A TVS X is *Locally Convex* if there exists a local basis of 0 that consists of convex sets.

Definition C.1.5. (Invariant Metric) A metric d on a TVS X is *invariant* if $d(x + h, y + h) = d(x, y)$

C.1.1 Banach and Frechet Topologies

Definition C.1.6. (Seminorm) A function $p : X \to \mathbb{R}$ is a seminorm if it satisfies the following:

1. $p(x) \geq 0, \forall x \in X$
2. $p(x + y) \leq p(x) + p(y)$
3. $p(\lambda x) = |\lambda| p(x)$ for $\lambda \in \mathbb{K}$.

Example C.1.7. (**Topology generated by a family of seminorms**) Let $\{p_i\}$ be a countable family of seminorms on a vector space X which *separates points*, i.e. if $p_i(x) = 0$ for all $i \in \mathbb{N}$, then $x = 0$. We make it a TVS in the following way: Let $U_{i,\delta}(z) := \{x \in X : p_i(x - z) < \delta\}$. This collection $\{U_{i,\delta}(z) : i \in \mathbb{N}, \delta > 0, z \in X\}$ forms a sub-basis, and generates the topology of X.

We can get a description of the topology so generated: $V_{\epsilon,k}(z) := \{y \in X : p_i(y-z) < \epsilon \ \forall i \leq k\}$ forms a basis for the TVS X as we vary $\epsilon > 0, k \in \mathbb{N}$ and $z \in X$. One can, in fact, show that the topology generated above is *metrizable* and the metric is given by

$$d(x,y) := \sum_{i=1}^{\infty} \frac{p_i(x-y)}{2^i(1+p_i(x-y))} \tag{C.1.1}$$

Let $B_\epsilon(0)$ be the open ball of radius ϵ at 0 coming form the metric above. Let ϵ be so that $\sum_{i=N}^{\infty} \frac{p_i(x)}{2^i(1+p_i(x))} < \frac{\epsilon}{2}$. Let x be so that $x \in V_{\delta=\frac{\epsilon}{2}(1-1/2^N),N}(0)$, i.e. $p_i(x) < \frac{\epsilon}{2}(1-1/2^N)$ for all $i \leq N$. Then we have

$$d(x,0) < \epsilon$$

which means that $V_{\delta,N}(0) \subset B_\epsilon(0)$. For the reverse inclusion: Let $d(x,0) = \sum_{i=1}^{\infty} \frac{p_i(x)}{2^i(p_i(x)+1)} < \delta$ which means for every $i \in \mathbb{N}$, $\frac{p_i(x)}{(p_i(x)+1)} < \delta 2^i$. Let δ be so that $2^K\delta < \epsilon/1+\epsilon$. Then for all $i \leq K$, $p_i(x) < \epsilon$ which implies $x \in \cap_{i=1}^{K} V_{\epsilon,i}(0)$ which is a basis element of the Frechet space. Hence, they form equivalent topologies.

Definition C.1.8. (Frechet Space) A TVS X is a Frechet Space if it is a complete metric space with an invariant metric, but not necessarily *normable*. Its topology comes from a countable family of seminorms $\{p_i\}$ as illustrated in C.1.7.

Theorem C.1.9. *(**Characterization of Frechet spaces**) A Hausdorff locally convex topological vector space is metrizable iff its topology is generated by a countable family of seminorms.*

Example C.1.10. (The Banach topology of $C^k(K)$) Let K be a compact set in $\mathbb{R}^n$. We define the norm on $C^k(K)$ as

$$||u||_{C^k(K)} := \sum_{|\alpha|=0}^{k} ||D^\alpha(u)||_{L^\infty(K)} \tag{C.1.2}$$

This is easily seen to be a genuine *norm*. Moreover, this space is complete with respect to this norm; See Theorem 8.2.3 in [BS11].

Remark C.1.11. **A word of caution:** *What does it mean for a function to be **differentiable** on a **compact set**?* We consider the following notion of differentiability: We say a function f is of class C^k on a compact set K if there is an open set $U \subseteq \mathbb{R}^n$ with $K \subset U$ and a function $\tilde{f} : U \to \mathbb{R}^m$ of class C^k such that $\tilde{f} = f$ when restricted on K.

Remark C.1.12. There is, however, a different notion by which we mean $C^k(\bar{\Omega})$. This is given below:

Definition C.1.13. **($C^k(\bar{\Omega})$)** Let Ω be a bounded domain. The collection of all functions f of class C^k on Ω whose derivatives $D^\alpha f$ extends continuously to the boundary of Ω.

Remark C.1.14. Note that, even though $\bar{\Omega}$ is compact for a bounded domain in $\mathbb{R}^n$, the two notions C.1.13 and C.1.11 are *not the same*. We require additional regularity on the set Ω for the two notions to be equal.

Frechet Topology of $C^k(\Omega)$

Let Ω be an open set in $\mathbb{R}^n$. We consider an exhaustion $\{K_i\}$ of Ω consisting of compact sets (e.g. $\{x \in \Omega : ||x|| \leq n \text{ and } \text{dist}(x, \Omega^c) \geq 1/n\}$). We define $p_i(u) := ||u||_{C^k(K_i)}$ (as defined in C.1.10). This forms a countable collection of semi-norms, and hence can be topologised as in C.1.7. A cauchy sequence is such that for every $i \in \mathbb{N}$,

$$p_i(u_n - u_m) \to 0$$

as $n, m \to \infty$. This means that in $C^k(K_i)$, $u_n|_{K_i}$ is cauchy and thus convergent (as $C^k(K_i)$ is Banach). Therefore, $C^k(\Omega)$ is a Frechet space.

Remark C.1.15. 1. It is easy to see that the collection $\{p_i\}$ separates points.

2. It should be noted that, the topology is independent of our choice of exhaustion $\{K_i\}$ as for any other exhaustion $\{F_j\}$, we see that they are *mutually dominant* (for every i, $\exists j(i)$ such that $K_i \subset F_{j(i)}$ and vice versa).

What it means to be Cauchy: A sequence u_n is cauchy in $C^k(\Omega)$ if given any neighbourhood $V_{\varepsilon,k}$ of 0, there exists n_0 such that $u_n - u_m \in V_{\varepsilon,k}$ for $n, m \geq n_0$. This can be recharacterised in the following way:

$$u_n \textit{ is Cauchy if and only if for every } i \in \mathbb{N},\ p_i(u_n - u_m) \to 0 \textit{ as } n, m \to \infty.$$

What it means to converge: A sequence u_n converges to u in $C^k(\Omega)$ if for every neighbourhood $V_{\varepsilon,s}$ of 0, there exists n_0 such that $u_n - u \in V_{\varepsilon,s}$. This can be recharacterised in the following way:

$$u_n \to u \text{ if and only if for every } i \in \mathbb{N},\ p_i(u_n - u) \to 0 \text{ as } n \to \infty.$$

Remark C.1.16. The above characterisation is true in general for any Frechet space.

Frechet Topology of $C^\infty(K)$

Let K be a compact subset of $\mathbb{R}^n$. Let

$$p_j(u) := ||u||_{C^{j-1}(K)}$$

This forms a countable family of seminorms that separate points, and hence $C^\infty(\Omega)$ is a Frechet space under this topology.

Frechet Topology of $C^\infty(\Omega)$

Let $\{K_i\}$ be an exhaustion of $\Omega \subset \mathbb{R}^n$. Let

$$p_{i,j}(u) := ||u||_{C^{j-1}(K_i)}$$

This yields a countable family of seminorms that separate points, and hence makes $C^\infty(\Omega)$ a Frechet space.

C.1.2 Inductive Limit Topologies

Definition C.1.17. (Inductive Limit of Frechet Spaces (LF spaces)) These are complete topological vector spaces (in the sense of cauchy nets or cauchy filters. cf.[SW99]) that are a union of of an increasing sequence E_n of Frechet spaces such that $E_n \hookrightarrow E_{n+1}$ is an isomorphism of topological vector spaces. $X = \cup_i E_i$ and we say a set E in X is open if and only if $E_i \cap E$ is open in E_i.

Remark C.1.18. 1. **LF spaces are not *in general* metrizable.**

2. The above point implies that it is ***not enough*** to consider *sequences* alone to try to charactetrise topologies, as we did in the case of Frechet spaces (which were metrizable).

LF Topology on $C_c(\Omega)$

Let $\{K_i\}$ be an exhaustion of Ω, and define

$$E_j := \{u \in C_c(\Omega) : \text{supp}(u) \subset K_j\}$$

We see that E_j is a vector space and we can give E_j a Banach space structure with the norm $||u||_{E_j} := ||u||_{L^\infty(K_j)}$. Moreover, $E_1 \subset E_2 \subset \cdots \subset \Omega$ and the inclusions are isomorphisms of Banach spaces.

LF Topology on $C_c^k(\Omega)$

Let $\{K_i\}$ be an exhaustion and

$$E_j := \{u \in C_c^k(\Omega) : \text{supp}(u) \subset K_j\}$$

This is a linear subspace of $C_c^k(\Omega)$, with the norm

$$||u||_{E_j} := ||u||_{C^k(K_j)} = \sum_{|\alpha|=1}^{k} ||D^\alpha u||_{L^\infty(K_j)}$$

which makes it a Banach Space. The inclusions $E_1 \hookrightarrow E_2 \hookrightarrow \cdots$ are also TVS isomorphisms, making $C_c^k(\Omega)$ an LF space.

LF Topology on $C_c^\infty(\Omega)$

Let

$$E_j := \{u \in C_c^\infty(\Omega) : \text{supp}(u) \subset K_j\}$$

and define

$$p_i^j(u) := ||u||_{C^i(K_j)}$$

This is a collection of semi-norms on E_j, making it a Frechet space. We then have

$$C_c^\infty(\Omega) = \cup_j E_j$$

making $C_c^\infty(\Omega)$ an LF space.

C.1.3 Criteria for convergence

In the Banach space setting, the criteria for convergence is quite simple: $u_n \to u$ if and only if $||u_n - u|| \to 0$.

The following proposition characterises convergence in Frechet spaces:

Proposition C.1.19. Let E be a Frechet space with a countable collection of seminorms $\{p_i\}$. A sequence u_n converges to u if and only if for each $i \in \mathbb{N}$,

$$p_i(u_n - u) \to u$$

as $n \to \infty$.

The following proposition characeterises convergence in LF spaces:

Proposition C.1.20. Let $E = \cup_i E_i$ be the inductive limit of Frechet spaces E_i with $E_1 \hookrightarrow E_2 \hookrightarrow \cdots \hookrightarrow E$. A sequence u_n converges to u in E if and only if there exists an integer $k \in \mathbb{N}$ such that

$$\{u_n\}_{n\in\mathbb{N}} \subset E_k,\ u \in E_k \text{ and } u_n \to u \in E_k (\text{in the Frechet sense})$$

Example C.1.21. Consider the space $C_c^k(\Omega)$ (with an exhaustion $\{K_j\}$ of Ω), which is an LF space, with $E_j := \{u \in C_c^k(\Omega) : \text{supp}(u) \subset K_j\}$, equipped with the norm

$$||u||_{E_j} := ||u||_{C^k(K_j)}$$

Let u_n be a sequence converging to u in $E = C_c^k(\Omega)$. This means that (from the characterisation in C.1.20) there exists a large enough q such that $\{u_n\}_{n\in\mathbb{N}} \subset E_q$, or rather, the supports of u_n are in K_q, and $u_n \to u$ in the norm sense, in E_q. This means,

$$||u_n - u||_{C^k(K_q)} = \sum_{|\alpha|=0}^{k} |D^\alpha(u_n) - D^\alpha(u)|_{L^\infty(K_q)} \to 0$$

which amounts to (since E_q is Banach) $D^\alpha(u)|_{K_q} \to D^\alpha u|_{K_q}$ uniformly in K_q for all $\alpha \in \{0, 1, \cdots, k\}$.

C.1.4 Continuous linear maps on LCTVS

Definition C.1.22. (Sequentially Continuous) A map $f : X \to Y$ between topological spaces is *sequentially continuous* if for every $x_n \to x$ in X, we have $f(x_n) \to f(x)$ in Y.

Proposition C.1.23. A map from a *first countable* topological space X to a topological space Y is continuous iff it is sequentially continuous.

We now describe what it means for a linear map between Frechet spaces to be continuous:

Continuity on Frechet space

Proposition C.1.24. Let E and F be Frechet spaces and $\{p_i\}_{i\in\mathbb{N}}$ and $\{q_i\}_{i\in\mathbb{N}}$ the associated countable family of seminorms. A linear map $T : E \to F$ is continuous if and only if for every $i \in \mathbb{N}$, there exists $j \in \mathbb{N}$, and a constant $C = C(i)$ such that $q_i(Tx) \leq Cp_j(x)$ for every $x \in E$.

Proposition C.1.25. Let E be a Frechet space with a countable family $\{p_i\}$ of seminorms generating its topology. A linear map $T : E \to \mathbb{K}$ is continuous iff for each integer $i \in \mathbb{N}$, there exists a constant $C = C(i)$ such that

$$|Tx| \leq Cp_i(x) \text{ for all } x \in E$$

Continuity on LF space

Let $E = \cup_i E_i$ where $\{E_i\}$ is an increasing collection of Frechet spaces, with $E_1 \hookrightarrow E_2 \hookrightarrow \cdots \hookrightarrow E$ being isomorphisms of TVS. We begin with an observation that the Frechet topology on E_i is the same as the topology induced from E. This amounts to the following:

Proposition C.1.26. Let E be an LF space with $E = \cup_i E_i$. Then any convex subset $U \subset E$ containing the origin is a neighbourhood of 0 if and only if for each $n \in \mathbb{N}$, $U \cap E_n$ is a neighbourhood of 0 in E_n.

We can now characterise continuous linear maps, with the following proposition:

Proposition C.1.27. Let E be an LF space with $E = cup_i E_i$, an increasing limit of a sequence of Frechet spaces. Let F be any LCTVS. A linear map $T : E \to F$ is continuous if and only if for each $n \in \mathbb{N}$,

$$T|_{E_n} \to F$$

is continuous in the Frechet sense.

C.2 Dual Spaces, Weak and Weak* Topologies of Normed Linear Spaces

C.2.1 Weak topology

Definition C.2.1. (Dual space X^*) Let X be a normed linear space over $\mathbb{K}$. The *topological dual space*(generally referred to as *the dual space*) is defined as the space of all *continuous linear functionals on* X.

We set up some notation: Let X be a normed linear space and X^* its topological dual. We can make a bilinear pairing

$$\langle , \rangle : X^* \times X \to \mathbb{K}$$

by

$$\langle f, x \rangle := f(x) \tag{C.2.1}$$

and we will choose to write $\langle f, x \rangle$ in place of $f(x)$.

Definition C.2.2. (The weak topology on X ($\sigma(X, X^*)$)) The weak topology on X, denoted $\sigma(X, X^*)$ is the *coarsest* topology that makes the collection $\{f : X \to \mathbb{K} : f \in X^*\}$ continuous.

Theorem C.2.3. ***(Basis for Weak Topology)*** *$\sigma(X, X^*)$ is generated by the collection*

$$\{\cap_{i=1}^n f_i^{-1}(U_i) : f_i \in X^*, U_i \text{ open in } \mathbb{K}, n \in \mathbb{N}\}$$

Proposition C.2.4. Let X be a normed linear space. $x_n \in X$ converges to $x \in X$ in the weak topology iff $\phi(x_n) \to \phi(x)$ for all $\phi \in X^*$

Convergence in the weak topology is called as *weak convergence.*
We gather a few facts about weak topology $\sigma(X, X^*)$ (without proof):

Lemma C.2.5. *The weak topology $\sigma(X, X^*)$ is Hausdorff.*

The proof comes from a direct application of the geometric form of the *Hahn-Banach Theorem.*

Proposition C.2.6. Let X be a NLS and $\sigma(X, X^*)$ the weak topology on X. This has a basis of the form $V(f_1, f_2, \cdots, f_k; x_0, \varepsilon)$ (where $\{f_i\}_{i=1}^k$ are elements of X^*, $x_0 \in X$, and $\varepsilon > 0$) given by

$$V(f_1, f_2, \cdots, f_k; x_0, \varepsilon) := \{x \in X : |\langle f, x - x_0 \rangle| < \varepsilon \text{ for all } i \in \{1, \cdots, k\}\}$$

Remark C.2.7. We make some important remarks:

1. Being weakly closed (complement of an open set in weak topology) *does not* imply being weakly *sequentially* closed (every weakly convergent sequence converges in the set itself).
2. In general, weakly closed implies weakly sequentially closed.
3. C being weakly closed *always* implies that C is strongly closed (recall that that the weak topology is indeed weaker than the strong norm topology), but *the converse is not always true.* For convex sets however, the coverse holds true.

Lemma C.2.8. *Let $C \subset X$ be a convex set that is strongly closed. Then C is weakly closed.*

The proof is again an application of the geomeric form of Hahn-Banach theorem.

Theorem C.2.9. ***Characterisation of weak convergence [Bre11]***

1. *$x_n \rightharpoonup x$ if and only if $f(x_n) \to f(x)$ for all $f \in X^*$.*
2. *Strong convergence in X implies weak convergence in $X, \sigma(X, X^*)$.*
3. *If $x_n \rightharpoonup x$, then $||x_n||$ is bounded and $||x|| \leq \liminf ||x_n||$.*
4. *If $x_n \rightharpoonup x$ and $f_n \to f$ strongly in X^*, then $\langle f_n, x_n \rangle \to \langle f, x \rangle$.*

C.2.2 Weak* topology

The J map

Consider for every x a map $J_x \in X^{**}$ whose action on a given linear functional $f \in X^*$ is given by

$$J_x(f) := \langle f, x \rangle \tag{C.2.2}$$

The map $x \mapsto J_x$ as a map from $X \to X^{**}$ is easily seen to be linear. Moreover, if $J_x = J_y$, then we would have $\langle f, x \rangle = \langle f, y \rangle$ for all $f \in X^*$ which implies $x = y$. Therefore, $J : X \to X^{**}$ is *injectlve*. Give X^{**} the operator norm topology, from which we get

$$||J_x|| := \sup_{f \in X^*, ||f||_{op}=1} |J_x(f)| = \sup_{f \in X^*, ||f||_{op}=1} |\langle f, x \rangle| = ||x||_X$$

from Hahn-Banach. Therefore, J is actually an *isometry* of Normed Linear Spaces, which means we can see X as an isometric subspace of X^{**}. We collect this result in the following theorem:

Theorem C.2.10. *Let X be a normed linear space and $J : X \to X^{**}$ be given by $J_x(f) := \langle f, x \rangle$ for all $f \in X^*$. Then $x \mapsto J_x$ is linear, continuous and an isometry of normed linear spaces.*

Definition C.2.11. (Weak* topology on $X^*(\sigma(X^*, X))$) We define the *weak** topology on X^* to be the coarsest topology on X^* that makes each $J_x : X^* \to \mathbb{K}$ continuous.

Remark C.2.12. This is easily seen to be *weaker* than the weak topology on X^*, as we require less maps to be continuous.

We now describe a useful basis for the weak* topology on X^*.

Proposition C.2.13. Let X be a NLS. X^* with the topology $\sigma(X^*, X)$ has a basis of the following form (with $x_1, x_2, \cdots, x_k \in X$, $\varepsilon > 0$ and $f_0 \in X^*$):

$$V(x_1, x_2, \cdots, x_k; \varepsilon, f_0) := \{f \in X^* : |\langle f - f_0, x_i \rangle| < \varepsilon, \text{ with } i \in \{1, \cdots, k\}\}$$

Notation for weak* convergence: When a sequence $f_n \in X^*$ converges to f in the *weak* topology*, we denote it by:

$$f_n \overset{*}{\rightharpoonup} f$$

Theorem C.2.14. ***Characterisations of weak* convergence[Bre11]***

1. *$f_n \overset{*}{\rightharpoonup} f$ iff $\langle f_n, x \rangle \to \langle f, x \rangle$ for all $x \in X$.*
2. *$f_n \to f$ in the operator norm implies $f_n \overset{*}{\rightharpoonup} f$ in $\sigma(X^*, X)$. Moreover, if $f_n \rightharpoonup f$ in $\sigma(X^*, X^{**})$, then $f \overset{*}{\rightharpoonup} f$ in $\sigma(X^*, X)$.*
3. *If $f_n \overset{*}{\rightharpoonup} f$, then $||f_n||$ is bounded and $||f|| \leq \liminf ||f_n||$.*
4. *If $f_n \overset{*}{\rightharpoonup} f$ and $x_x \to x$ strongly, then $\langle f_n, x_n \rangle \to \langle f, x \rangle$.*

Bibliography

[Bre11] Haim Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. New York: Springer, 2011. ISBN: 978-0-387-70913-0.

[BS11] Robert G. Bartle and Donald R. Sherbert. *Introduction to Real Analysis*. 4th ed. Hoboken, NJ: John Wiley & Sons, 2011. ISBN: 978-0-471-43331-6.

[CDK12] Gyula Csató, Bernard Dacorogna, and Olivier Kneuss. *The Pullback Equation for Differential Forms*. Vol. 83. Progress in Nonlinear Differential Equations and Their Applications. New York: Birkhäuser, 2012. ISBN: 978-0-8176-8312-2. DOI: 10.1007/978-0-8176-8313-9.

[CDS18] G. Csato, B. Dacorogna, and S. Sil. "On the best constant in Gaffney inequality". In: *Journal of Functional Analysis* 274.2 (Jan. 2018), pp. 461–503. ISSN: 0022-1236. DOI: 10.1016/j.jfa.2017.10.006. URL: http://dx.doi.org/10.1016/j.jfa.2017.10.006.

[Jos17] Jürgen Jost. *Riemannian Geometry and Geometric Analysis*. 7th ed. Universitext. Cham: Springer, 2017. ISBN: 978-3-319-61859-3.

[Rud91] Walter Rudin. *Functional Analysis*. 2nd ed. International Series in Pure and Applied Mathematics. New York: McGraw-Hill, 1991. ISBN: 0-07-054236-8.

[SW99] Helmut H. Schaefer and Manfred P. Wolff. *Topological Vector Spaces*. 2nd ed. Vol. 3. Graduate Texts in Mathematics. New York: Springer, 1999. ISBN: 978-0-387-98726-2.